Global Memory Tech

Weekly theme: Micron results implications for Asian memory chipmakers

Industry Overview

Five implications from Micron's results/guidance

We present five implications for Asian memory chipmakers following Micron's upbeat results/guidance: (1) super-cycle or chip shortage likely to continue into 2027 or even 2030; (2) more LTAs (long-term agreements) with big tech companies and OEMs, which could make the industry less cyclical and highly profitable (such as Taiwan's proven foundry); (3) not easy to build new shell fabs (high construction cost, local government regulation, power/water supply issues, etc.); (4) limited wafer capacity expansion even in 2028 (lots of clean rooms needed for old fab upgrades, longer manufacturing cycles, larger sized front-end and even back-end equipment, low yields in producing more advanced chips, high trade ratio for HBM vs conventional DRAM); and (5) significantly increasing FCF despite more-than-doubled capex spend vs normal cycle. Micron also pointed to meaningful HBM4 sales (US$1bn so far); new fab construction / operations across the globe – the US (Boise; New York; Virginia), Taiwan (Tongluo), Singapore, and Japan; as well as large EUV orders. Our check also suggests that all major Asian memory chipmakers also agree on Micron's bullish memory industry views.

Fine-tune global memory forecasts

We also reflect Micron's results (upbeat ASP for May'26-end quarter) and guidance (just ~20% revenue growth QoQ in Aug-end quarter vs +75%/+74% in Feb/May-end quarters) in our industry model. We believe ASP will no longer rise strongly if LTA-based sales increase. We also learned of stronger demand/production for HBM4, HBM4e, SOCAMM, LPDDR5, GDDR7, eSSD, etc. This could lead to higher bit growth in 2H26 and 2027 (but still sub-20% YoY, in our view). Overall, we tweak up our 2026-28 global DRAM/NAND sales forecasts by 2-4%, but the revisions are mostly based on upbeat 2Q26 ASP (offsetting muted 3Q or 2H ASP due to LTAs) and slightly higher volume increase for 2H26 and 2027-28. We also note Micron's revenue guidance (US$50bn for Aug-end quarter; annual run rate US$200bn; implied memory industry market size US$1.0tn with Micron's 20%+/- market share) is consistent with our industry view (3Q global DRAM/NAND sales total US$257bn; annual run rate also US$1.0tn). Korea's semis exports (mostly memory) remained strong in June (+188% YoY for the first 20-day period). DRAM spot price has also risen for five consecutive weeks (up 20%) vs softened NAND (down 5%).

Memory chipmakers unlikely to face severe price competition

Micron and other large memory chipmakers are increasingly using LTAs to fix ASPs; their chip production will also be more high-end centric (HBM, SOCAMM, LPDDR5). This could unlock new growth opportunities for memory chipmakers without severe price competition. We note, for example, that Nanya Tech's DRAM sales (mostly old/legacy DRAM) are still mainly based on monthly/quarterly negotiated ASPs (very low exposure to LTAs).

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Refer to important disclosures on page 28 to 29.

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Exhibit 1: Global memory forecasts – super-cycle should continue into 2027 even after quadrupling in 2026

BofA global memory forecasts

YoY	24	25	26E	27E
DRAM sales	86%	52%	316%	43%
NAND sales	84%	4%	295%	29%
DRAM ASP	62%	29%	242%	23%
NAND ASP	65%	-8%	234%	8%
DRAM bit	15%	18%	22%	17%
NAND bit	11%	13%	18%	20%
DRAM capex	49%	43%	65%	21%
NAND capex	4%	-5%	56%	16%
DRAM capa	7%	8%	11%	9%
NAND capa	-2%	-7%	5%	4%

Source: BofA Global Research estimates

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Exhibit 2: DDR5 and DDR4 price up in June, while NAND prices softened

Spot-market prices among DRAM and NAND

US$	Current	WoW	QoQ	YoY
DRAM spot				
16Gb DDR5	46.7	2%	25%	672%
16Gb DDR4	71.6	4%	-4%	737%
8Gb DDR4	67.8	1%	-4%	721%
NAND spot				
1Tb wafer	24.5	-2%	-12%	383%
512Gb wafer	20.1	-2%	-12%	648%
256Gb wafer	10.2	-2%	-9%	581%

Source: DRAMeXchange

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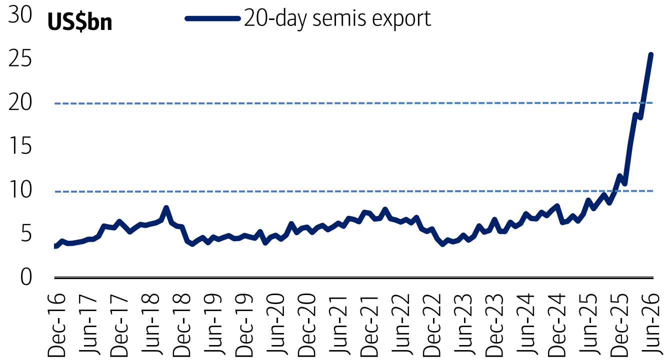
For abbreviations, see glossary at the end of this report.

Korea's exports and Micron's results

Exhibit 3: Strong MoM growth in Jun (US$25.5bn; +16% MoM); more than 3x higher than 2025 year-average level

Korea semis exports (mostly memory) – first 20 days of month (US$bn)



Source: MoTIR

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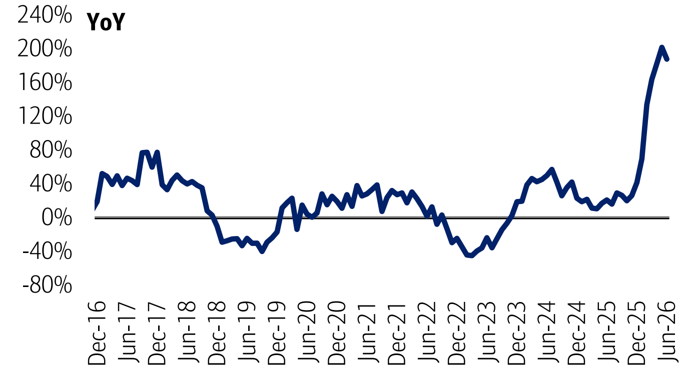
Exhibit 4: Up 188% YoY in Jun; already five consecutive months of triple-digit growth

Korea semis exports – YoY in first 20 days of month



Source: MoTIR

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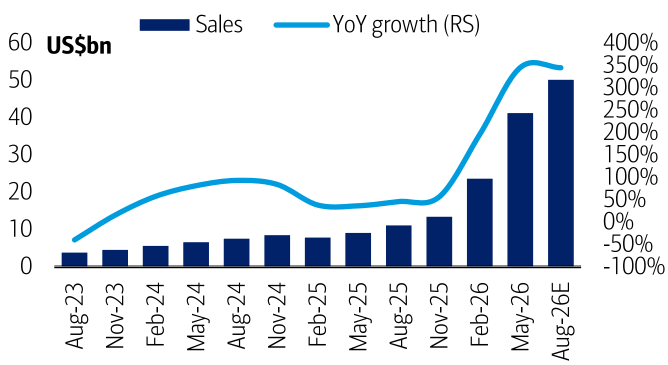
Exhibit 5: Record-high sales in May-Q (US$41bn; +346% YoY); mgmt also provided solid guidance for Aug-Q (US$50bn), which implies +21% QoQ growth

Micron – Sales and YoY growth trend



Source: Company, BofA Global Research estimates

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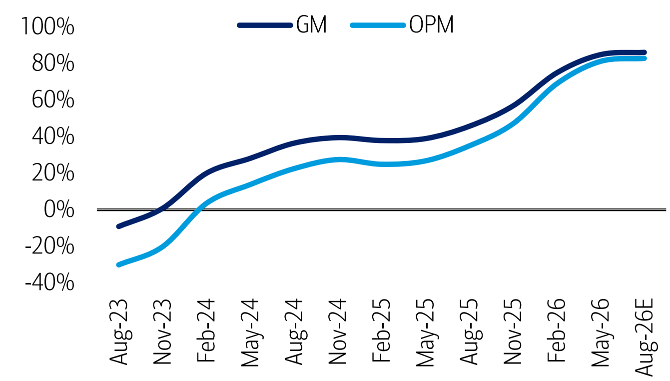
Exhibit 6: Significant margin enhancement in May-Q (GM 85%, OPM 81%); mgmt. expects high-margin profile to continue

Micron – Gross/OP margin trend



Source: Company, BofA Global Research estimates

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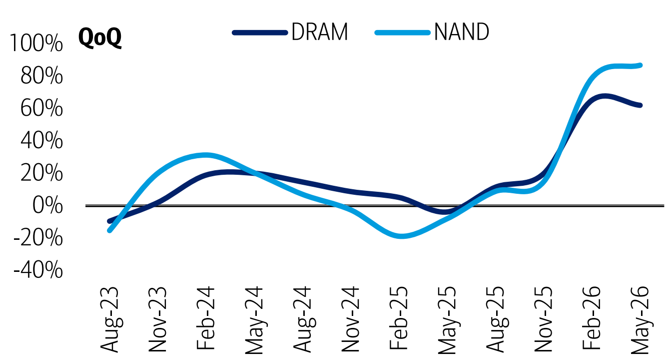
Exhibit 7: Robust price hike observed in May-Q (DRAM up low-60% QoQ and NAND up mid-80%)

Micron – DRAM and NAND ASP trend (QoQ basis)



Source: Company, BofA Global Research estimates

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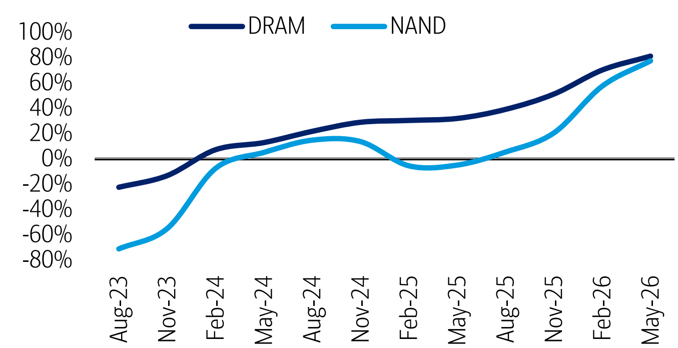
Exhibit 8: Record-high margins in May-Q (DRAM 81%, NAND 78%)

Micron – DRAM and NAND OP margin trend



Source: Company, BofA Global Research estimates

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Global memory forecasts

Exhibit 9: We expect global DRAM revenue to nearly quadruple (+316% YoY) in 2026E, building on the strong growth trajectory of 86%/52% in 2024/25, primarily driven by a sharp ~3x rebound in ASPs (+242% YoY). NAND revenue is similarly forecast to increase nearly fourfold (+295% YoY) in 2026E, following modest low-single-digit growth in 2025, supported by a robust ~2.3x expansion in ASPs (+234% YoY).

Global memory forecast summary – top-down approach

	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
Revenue US$bn																
DRAM	83.9	134.6	162.1	176.6	185.5	193.5	206.3	213.9	92.1	77.7	47.3	87.9	133.8	557.2	799.1	870.9
NAND	45.3	78.9	94.7	100.6	99.9	101.3	104.9	106.9	73.3	67.3	42.4	78.1	81.0	319.6	413.0	414.1
DRAM+NAND total	**129.2**	**213.6**	**256.8**	**277.2**	**285.4**	**294.8**	**311.2**	**320.7**	**165.4**	**144.9**	**89.7**	**166.0**	**214.8**	**876.8**	**1,212.2**	**1,285.1**
% QoQ																
DRAM	78%	60%	20%	9%	5%	4%	7%	4%								
NAND	71%	74%	20%	6%	-1%	1%	4%	2%								
DRAM/NAND total	**75%**	**65%**	**20%**	**8%**	**3%**	**3%**	**6%**	**3%**								
% YoY																
DRAM	258%	378%	362%	275%	121%	44%	27%	21%	41%	-16%	-39%	86%	52%	316%	43%	9%
NAND	196%	332%	352%	280%	121%	28%	11%	6%	25%	-8%	-37%	84%	4%	295%	29%	0%
DRAM/NAND total	**234%**	**360%**	**359%**	**276%**	**121%**	**38%**	**21%**	**16%**	**34%**	**-12%**	**-38%**	**85%**	**29%**	**308%**	**38%**	**6%**
Blended ASP US$/unit																
DRAM 8Gb equiv	8.1	12.4	14.4	15.5	15.8	15.8	15.6	15.3	3.8	3.2	1.8	2.9	3.7	12.7	15.6	14.4
NAND 256Gb equiv	5.0	8.3	9.4	9.5	9.3	9.0	8.6	8.3	3.4	3.0	1.6	2.6	2.4	8.1	8.8	7.6
% QoQ																
DRAM ASP	73%	53%	17%	7%	2%	0%	-2%	-2%								
NAND ASP	75%	65%	13%	1%	-2%	-3%	-5%	-3%								
% YoY																
DRAM ASP	155%	283%	304%	231%	95%	28%	8%	-1%	14%	-16%	-45%	62%	29%	242%	23%	-8%
NAND ASP	115%	281%	306%	230%	86%	9%	-9%	-12%	-9%	-13%	-46%	65%	-8%	234%	8%	-13%
Shipments bn units																
DRAM 8Gb equiv	10.4	10.9	11.2	11.4	11.8	12.2	13.3	14.0	24.1	24.2	26.7	30.6	36.0	43.9	51.2	60.5
NAND 256Gb equiv	9.0	9.5	10.1	10.6	10.7	11.2	12.2	12.8	21.3	22.6	26.6	29.5	33.2	39.3	47.0	54.3
% QoQ																
DRAM bit growth	3%	5%	3%	2%	3%	4%	8%	5%								
NAND bit growth	-2%	5%	6%	5%	1%	5%	9%	5%								
% YoY																
DRAM bit growth	41%	25%	14%	13%	13%	12%	18%	22%	23%	0%	10%	15%	18%	22%	17%	18%
NAND bit growth	38%	13%	11%	15%	19%	18%	21%	21%	38%	6%	18%	11%	13%	18%	20%	16%
Capacity 300mm k wpm																
DRAM wafer capacity	1,988	2,036	2,091	2,150	2,182	2,224	2,264	2,299	1,481	1,561	1,628	1,734	1,867	2,066	2,242	2,379
NAND wafer capacity	1,695	1,722	1,736	1,750	1,767	1,784	1,801	1,834	1,681	1,783	1,820	1,781	1,651	1,726	1,797	1,878
DRAM+NAND total	**3,683**	**3,758**	**3,827**	**3,900**	**3,949**	**4,008**	**4,065**	**4,133**	**3,162**	**3,344**	**3,447**	**3,515**	**3,517**	**3,792**	**4,039**	**4,256**
% QoQ																
DRAM capacity total	3%	2%	3%	3%	1%	2%	2%	2%								
NAND capacity total	1%	2%	1%	1%	1%	1%	1%	2%								
DRAM/NAND total	2%	2%	2%	2%	1%	1%	1%	2%								
% YoY																
DRAM capacity	9%	11%	12%	11%	10%	9%	8%	7%	8%	5%	4%	7%	8%	11%	9%	6%
NAND capacity	2%	6%	6%	4%	4%	4%	4%	5%	1%	6%	2%	-2%	-7%	5%	4%	5%
DRAM+NAND total	**5%**	**9%**	**9%**	**8%**	**7%**	**7%**	**6%**	**6%**	**4%**	**6%**	**3%**	**2%**	**0%**	**8%**	**7%**	**5%**
Capex spending																
DRAM	23.2	22.3	21.4	22.3	24.8	25.9	28.0	29.1	27.1	31.0	25.2	37.7	53.9	89.1	107.9	112.3
NAND	6.9	7.2	7.5	8.4	9.1	8.7	8.4	8.7	27.9	29.0	19.7	20.4	19.3	30.1	34.8	36.3
Total (DRAM+NAND)	**30.1**	**29.5**	**28.9**	**30.7**	**33.9**	**34.6**	**36.4**	**37.8**	**55.0**	**60.0**	**45.0**	**58.2**	**73.2**	**119.2**	**142.7**	**148.6**
QoQ/YoY in capex																
DRAM	59%	-4%	-4%	4%	11%	4%	8%	4%	27%	14%	-19%	49%	43%	65%	21%	4%
NAND	28%	4%	4%	12%	7%	-4%	-4%	4%	24%	4%	-32%	4%	-5%	56%	16%	4%
Total (DRAM+NAND)	**51%**	**-2%**	**-2%**	**6%**	**10%**	**2%**	**5%**	**4%**	**25%**	**9%**	**-25%**	**29%**	**26%**	**63%**	**20%**	**4%**

Source: Companies' reports, BofA Global Research estimates

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Exhibit 10: We slightly raise our 2026–28 DRAM revenue forecasts by 2–4%, primarily reflecting higher ASP assumptions (at ~$13/$16/$14 per 8Gb equivalent in 2026/27/28). NAND revenue estimates are also increased by 4% for 2027–28, driven by stronger ASP expectations (~$8–9 per 256Gb equivalent) amid the recent pricing upcycle.

Global memory forecast revisions – top-down analysis

	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
DRAM (8Gb equiv)																
Sales (US$bn)																
New	83.9	134.6	162.1	176.6	185.5	193.5	206.3	213.9	92.1	77.7	47.3	87.9	133.8	557.2	799.1	870.9
Old	83.9	128.9	159.2	175.0	180.3	188.1	201.7	209.1	92.1	77.7	47.3	87.9	133.8	547.0	779.1	838.0
Diff	nm	4.4%	1.8%	0.9%	2.9%	2.9%	2.3%	2.3%	nm	nm	nm	nm	nm	1.9%	2.6%	3.9%
Shipments (bn units)																
New	10.4	10.9	11.2	11.4	11.8	12.2	13.3	14.0	24.1	24.2	26.7	30.6	36.0	43.9	51.2	60.5
Old	10.4	11.0	11.3	11.5	11.7	12.1	13.1	13.8	24.1	24.2	26.7	30.6	36.0	44.3	50.8	58.8
Diff	nm	-1.0%	-1.0%	-1.0%	0.9%	0.9%	0.9%	0.9%	nm	nm	nm	nm	nm	-0.8%	0.9%	2.9%
Bit growth - sequential																
New	2.8%	5.0%	3.0%	1.8%	3.0%	4.1%	8.3%	5.3%	23.3%	0.3%	10.3%	14.8%	17.6%	21.9%	16.7%	18.1%
Old	2.8%	6.1%	3.0%	1.8%	1.0%	4.1%	8.3%	5.3%	23.3%	0.3%	10.3%	14.8%	17.6%	22.9%	14.7%	15.8%
Diff	nm	-1.1%	0.0%	0.0%	2.0%	0.0%	0.0%	0.0%	nm	nm	nm	nm	nm	-1.0%	2.0%	2.3%
ASP chg - sequential																
New	73.2%	52.9%	16.9%	7.0%	2.0%	0.2%	-1.6%	-1.6%	14.2%	-16.0%	-44.8%	61.8%	29.4%	241.6%	22.9%	-7.7%
Old	73.2%	44.9%	19.9%	8.0%	2.0%	0.2%	-1.0%	-1.5%	14.2%	-16.0%	-44.8%	61.8%	29.4%	232.7%	24.2%	-7.1%
Diff	nm	8.0%	-3.0%	-1.0%	0.0%	0.0%	-0.6%	0.0%	nm	nm	nm	nm	nm	8.9%	-1.3%	-0.6%
Blended ASP (US$)																
New	8.1	12.4	14.4	15.5	15.8	15.8	15.6	15.3	3.8	3.2	1.8	2.9	3.7	12.7	15.6	14.4
Old	8.1	11.7	14.0	15.2	15.5	15.5	15.3	15.1	3.8	3.2	1.8	2.9	3.7	12.4	15.3	14.2
Diff	nm	5.5%	2.9%	1.9%	1.9%	1.9%	1.4%	1.4%	nm	nm	nm	nm	nm	2.7%	1.6%	1.0%
NAND (256Gb equiv)																
Sales (US$bn)																
New	45.3	78.9	94.7	100.6	99.9	101.3	104.9	106.9	73.3	67.3	42.4	78.1	81.0	319.6	413.0	414.1
Old	45.3	76.2	96.5	101.5	97.3	96.9	100.3	102.2	73.3	67.3	42.4	78.1	81.0	319.5	396.6	397.1
Diff	nm	3.6%	-1.8%	-0.9%	2.7%	4.6%	4.6%	4.6%	nm	nm	nm	nm	nm	0.0%	4.1%	4.3%
Shipments (bn units)																
New	9.0	9.5	10.1	10.6	10.7	11.2	12.2	12.8	21.3	22.6	26.6	29.5	33.2	39.3	47.0	54.3
Old	9.0	9.5	10.5	11.0	10.8	11.0	12.0	12.6	21.3	22.6	26.6	29.5	33.2	40.0	46.5	53.5
Diff	nm	0.2%	-3.4%	-3.4%	-0.3%	1.5%	1.5%	1.5%	nm	nm	nm	nm	nm	-1.8%	1.1%	1.7%
Bit growth - sequential																
New	-2.2%	5.4%	6.2%	5.2%	0.9%	4.6%	9.0%	5.0%	37.8%	6.1%	17.6%	11.2%	12.5%	18.2%	19.7%	15.6%
Old	-2.2%	5.2%	10.1%	5.2%	-2.2%	2.7%	9.0%	5.0%	37.8%	6.1%	17.6%	11.2%	12.5%	20.3%	16.3%	15.0%
Diff	nm	0.2%	-3.9%	0.0%	3.1%	1.9%	0.0%	0.0%	nm	nm	nm	nm	nm	-2.1%	3.4%	0.6%
ASP chg - sequential																
New	74.7%	65.4%	13.0%	1.0%	-1.6%	-3.1%	-5.0%	-3.0%	-9.1%	-13.5%	-46.4%	65.5%	-7.8%	234.0%	7.9%	-13.3%
Old	74.7%	60.0%	15.0%	0.0%	-2.0%	-3.1%	-5.0%	-3.0%	-9.1%	-13.5%	-46.4%	65.5%	-7.8%	228.0%	6.7%	-12.9%
Diff	nm	5.4%	-2.0%	1.0%	0.4%	0.0%	0.0%	0.0%	nm	nm	nm	nm	nm	6.0%	1.2%	-0.3%
Blended ASP (US$)																
New	5.0	8.3	9.4	9.5	9.3	9.0	8.6	8.3	3.4	3.0	1.6	2.6	2.4	8.1	8.8	7.6
Old	5.0	8.0	9.2	9.2	9.0	8.8	8.3	8.1	3.4	3.0	1.6	2.6	2.4	8.0	8.5	7.4
Diff	nm	3.4%	1.6%	2.6%	3.0%	3.0%	3.0%	3.0%	nm	nm	nm	nm	nm	1.8%	3.0%	2.6%

Source: BofA Global Research estimates, industry data. Chg = change. Equiv = equivalent

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Exhibit 11: Servers—especially AI systems using HBM—now make up over half of total DRAM demand, mainly due to higher memory per system
Global DRAM demand mix by application

(8Gb equiv, mn units)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2022	2023	2024	2025E	2026E	2027E	2028E
Supply-to-demand															
Sufficiency (supply-to-demand)	**64.1%**	**75.0%**	**88.0%**	**96.0%**	**101.0%**	**100.0%**	**101.0%**	**101.8%**	**108.5%**	**100.6%**	**94.3%**	**92.1%**	**79.3%**	**101.0%**	**101.8%**
Supply	10,375	10,894	11,220	11,422	11,765	12,247	13,264	13,968	24,171	26,670	30,628	36,023	43,912	51,245	60,514
Demand	16,196	14,525	12,751	11,898	11,649	12,247	13,133	13,728	22,270	26,514	32,491	39,106	55,370	50,757	59,462
Wafer capacity (12" equiv k wpm)	**1,988**	**2,036**	**2,091**	**2,150**	**2,182**	**2,224**	**2,264**	**2,299**	**1,561**	**1,628**	**1,734**	**1,867**	**2,066**	**2,242**	**2,379**
Revenue (US$bn)	**83.9**	**134.6**	**162.1**	**176.6**	**185.5**	**193.5**	**206.3**	**213.9**	**77.7**	**47.3**	**87.9**	**133.8**	**557.2**	**799.1**	**870.9**
ASP (US$)	**8.1**	**12.4**	**14.4**	**15.5**	**15.8**	**15.8**	**15.6**	**15.3**	**3.2**	**1.8**	**2.9**	**3.7**	**12.7**	**15.6**	**14.4**
Demand by application (mn units)															
Server	**5,468**	**5,583**	**6,094**	**6,585**	**6,452**	**6,646**	**7,599**	**7,596**	**9,731**	**9,022**	**13,217**	**18,459**	**23,730**	**28,293**	**34,504**
PC	**1,028**	**869**	**864**	**803**	**707**	**747**	**838**	**913**	**3,574**	**3,418**	**3,922**	**4,713**	**3,564**	**3,205**	**3,694**
Smartphone	**2,844**	**2,481**	**2,430**	**2,526**	**2,296**	**2,366**	**2,558**	**2,794**	**7,388**	**8,045**	**10,255**	**12,149**	**10,281**	**10,014**	**11,586**
Tablet	233	194	183	196	184	188	187	221	846	786	929	1,052	805	781	908
TV - DRAM embedded	221	206	238	297	251	237	273	341	586	635	728	839	962	1,102	1,262
Game - DRAM embedded	125	114	136	206	135	124	156	221	454	459	499	535	581	636	680
Auto - DRAM embedded	105	113	125	132	135	145	161	170	168	219	288	370	475	611	785
Total demand	**16,196**	**14,525**	**12,751**	**11,898**	**11,649**	**12,247**	**13,133**	**13,728**	**22,270**	**26,514**	**32,491**	**39,106**	**55,370**	**50,757**	**59,462**
Demand mix (% to total bits)															
Server	**34%**	**38%**	**48%**	**55%**	**55%**	**54%**	**58%**	**55%**	**44%**	**34%**	**41%**	**47%**	**43%**	**56%**	**58%**
PC	**6%**	**6%**	**7%**	**7%**	**6%**	**6%**	**6%**	**7%**	**16%**	**13%**	**12%**	**12%**	**6%**	**6%**	**6%**
Smartphone	**18%**	**17%**	**19%**	**21%**	**20%**	**19%**	**19%**	**20%**	**33%**	**30%**	**32%**	**31%**	**19%**	**20%**	**19%**
Tablet	1%	1%	1%	2%	2%	2%	1%	2%	4%	3%	3%	3%	1%	2%	2%
TV - DRAM embedded	1%	1%	2%	2%	2%	2%	2%	2%	3%	2%	2%	2%	2%	2%	2%
Game - DRAM embedded	1%	1%	1%	2%	1%	1%	1%	2%	2%	2%	2%	1%	1%	1%	1%
Auto - DRAM embedded	0.6%	0.8%	1.0%	1.1%	1.2%	1.2%	1.2%	1.2%	0.8%	0.8%	0.9%	0.9%	0.9%	1.2%	1.3%
Total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
DRAM content per product (GB)															
Server	**1,148**	**1,159**	**1,182**	**1,194**	**1,206**	**1,230**	**1,267**	**1,293**	**655**	**736**	**914**	**1,066**	**1,172**	**1,251**	**1,383**
PC	**16.9**	**15.7**	**15.4**	**14.2**	**14.3**	**14.6**	**14.9**	**15.1**	**12.2**	**13.5**	**15.3**	**17.1**	**15.6**	**14.8**	**16.2**
Smartphone	**10.4**	**10.2**	**9.8**	**9.7**	**9.8**	**10.0**	**10.3**	**10.8**	**6.2**	**7.0**	**8.4**	**9.8**	**10.0**	**10.3**	**11.3**
Tablet	6.8	6.4	6.2	6.3	6.4	6.4	6.6	7.0	5.2	5.7	6.3	7.0	6.5	6.6	7.3
TV - DRAM embedded	4.5	4.6	4.7	4.8	5.1	5.2	5.3	5.4	2.9	3.2	3.6	4.1	4.7	5.3	5.9
Game - DRAM embedded	13.1	13.3	13.5	13.6	13.9	14.2	14.3	14.4	10.6	11.2	11.9	12.6	13.4	14.2	15.1
Auto - DRAM embedded	5.2	5.5	5.8	5.9	6.3	6.7	6.9	7.1	2.4	3.0	3.9	4.7	5.6	6.8	8.1
DRAM-based products (mn units)															
Server	**5**	**5**	**5**	**6**	**5**	**5**	**6**	**6**	**15**	**12**	**14**	**17**	**20**	**23**	**25**
PC	**61**	**55**	**56**	**57**	**49**	**51**	**56**	**61**	**294**	**253**	**256**	**275**	**228**	**217**	**228**
Smartphone	**275**	**244**	**247**	**259**	**233**	**236**	**247**	**260**	**1,198**	**1,152**	**1,218**	**1,237**	**1,025**	**976**	**1,028**
Tablet	34	31	29	31	29	29	28	32	163	138	147	151	125	118	124
TV - DRAM embedded	49	45	50	61	50	46	51	63	203	199	201	203	206	209	213
Game - DRAM embedded	10	9	10	15	10	9	11	15	43	41	42	42	43	45	45
Auto - DRAM embedded	20	20	22	22	21	22	23	24	69	72	75	79	85	90	96
%QoQ in DRAM-based products															
Server	**0%**	**1%**	**7%**	**7%**	**-3%**	**1%**	**11%**	**-2%**	**10%**	**-17%**	**18%**	**20%**	**17%**	**12%**	**10%**
PC	**-17%**	**-9%**	**1%**	**1%**	**-13%**	**3%**	**10%**	**8%**	**-16%**	**-14%**	**1%**	**8%**	**-17%**	**-5%**	**5%**
Smartphone	**-15%**	**-11%**	**1%**	**5%**	**-10%**	**1%**	**5%**	**5%**	**-12%**	**-4%**	**6%**	**2%**	**-17%**	**-5%**	**5%**
Tablet	-20%	-10%	-4%	6%	-7%	1%	-4%	13%	-12%	-16%	7%	3%	-17%	-5%	5%
TV - DRAM embedded	-19%	-9%	12%	22%	-19%	-8%	12%	22%	-5%	-2%	1%	1%	1%	2%	2%
Game - DRAM embedded	-35%	-11%	19%	50%	-36%	-10%	25%	40%	-10%	-5%	3%	1%	2%	3%	1%
Auto - DRAM embedded	-4%	2%	7%	3%	-4%	2%	7%	3%	0%	5%	3%	6%	7%	7%	7%

Source: BofA Global Research estimates, industry data. Chg = change. Equiv = equivalent

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Exhibit 12: SSD—especially in data center and AI applications—now accounts for over half of total NAND demand, driven by higher storage capacity per system

Global NAND demand mix by application

(256Gb equiv, mn units)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2022	2023	2024	2025E	2026E	2027E	2028E
Supply-to-demand															
Sufficiency (supply-to-demand)	**65.1%**	**69.2%**	**92.4%**	**99.9%**	**101.0%**	**103.0%**	**104.0%**	**102.7%**	**108.0%**	**101.8%**	**94.9%**	**97.7%**	**79.8%**	**102.7%**	**102.2%**
Supply	9,022	9,510	10,099	10,624	10,720	11,217	12,227	12,838	22,576	26,558	29,528	33,222	39,256	47,003	54,350
Demand	13,853	13,742	10,930	10,639	10,614	10,891	11,757	12,498	20,910	26,097	31,128	34,004	49,164	45,759	53,172
Wafer capacity (12" equiv k wpm)	**1,695**	**1,722**	**1,736**	**1,750**	**1,767**	**1,784**	**1,801**	**1,834**	**1,783**	**1,820**	**1,781**	**1,651**	**1,726**	**1,797**	**1,878**
Revenue (US$bn)	**45.3**	**78.9**	**94.7**	**100.6**	**99.9**	**101.3**	**104.9**	**106.9**	**67.3**	**42.4**	**78.1**	**81.0**	**319.6**	**413.0**	**414.1**
ASP (US$)	**5.0**	**8.3**	**9.4**	**9.5**	**9.3**	**9.0**	**8.6**	**8.3**	**3.0**	**1.6**	**2.6**	**2.4**	**8.1**	**8.8**	**7.6**
Demand by application (mn units)															
Smartphones	**2,808**	**2,474**	**2,449**	**2,623**	**2,408**	**2,481**	**2,683**	**2,902**	**6,821**	**7,724**	**9,792**	**11,965**	**10,355**	**10,473**	**12,467**
SSDs	**3,932**	**4,033**	**4,511**	**4,928**	**4,392**	**4,651**	**5,379**	**5,860**	**9,520**	**9,715**	**12,587**	**15,738**	**17,405**	**20,282**	**23,880**
Notebook	*1,129*	*1,013*	*1,061*	*1,072*	*961*	*1,036*	*1,178*	*1,316*	*3,172*	*3,275*	*3,991*	*4,934*	*4,275*	*4,491*	*5,325*
Desktop	*431*	*437*	*482*	*502*	*445*	*455*	*502*	*523*	*990*	*1,031*	*1,271*	*1,708*	*1,853*	*1,925*	*2,004*
Server	*1,929*	*2,105*	*2,428*	*2,749*	*2,467*	*2,617*	*3,085*	*3,333*	*4,256*	*4,432*	*6,057*	*7,335*	*9,211*	*11,502*	*13,866*
Spot & inventories	*443*	*477*	*540*	*606*	*519*	*542*	*614*	*689*	*1,101*	*978*	*1,268*	*1,760*	*2,066*	*2,363*	*2,686*
Tablets	393	335	312	335	311	321	316	370	1,134	1,155	1,483	1,812	1,374	1,319	1,509
Game - NAND embedded	216	196	235	357	235	216	273	386	736	760	843	913	1,004	1,111	1,202
Autos - NAND embedded	55	58	66	72	72	77	87	94	80	107	147	191	251	330	434
Others & spot adj	6,449	6,645	3,356	2,324	3,195	3,145	3,019	2,885	2,620	6,636	6,277	3,385	18,775	12,245	13,679
NAND demand total	**13,853**	**13,742**	**10,930**	**10,639**	**10,614**	**10,891**	**11,757**	**12,498**	**20,910**	**26,097**	**31,128**	**34,004**	**49,164**	**45,759**	**53,172**
Demand mix (% to total bits)															
Smartphones	**20%**	**18%**	**22%**	**25%**	**23%**	**23%**	**23%**	**23%**	**33%**	**30%**	**31%**	**35%**	**21%**	**23%**	**23%**
SSDs	**28%**	**29%**	**41%**	**46%**	**41%**	**43%**	**46%**	**47%**	**46%**	**37%**	**40%**	**46%**	**35%**	**44%**	**45%**
Notebook	*8%*	*7%*	*10%*	*10%*	*9%*	*10%*	*10%*	*11%*	*15%*	*13%*	*13%*	*15%*	*9%*	*10%*	*10%*
Desktop	*3%*	*3%*	*4%*	*5%*	*4%*	*4%*	*4%*	*4%*	*5%*	*4%*	*4%*	*5%*	*4%*	*4%*	*4%*
Server	*14%*	*15%*	*22%*	*26%*	*23%*	*24%*	*26%*	*27%*	*20%*	*17%*	*19%*	*22%*	*19%*	*25%*	*26%*
Spot & inventories	*3%*	*3%*	*5%*	*6%*	*5%*	*5%*	*5%*	*6%*	*5%*	*4%*	*4%*	*5%*	*4%*	*5%*	*5%*
Tablets	3%	2%	3%	3%	3%	3%	3%	3%	5%	4%	5%	5%	3%	3%	3%
Game - NAND embedded	2%	1%	2%	3%	2%	2%	2%	3%	4%	3%	3%	3%	2%	2%	2%
Autos - NAND embedded	0%	0%	1%	1%	1%	1%	1%	1%	0%	0%	0%	1%	1%	1%	1%
Total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
NAND content per product (GB)															
Smartphones	**327**	**324**	**318**	**324**	**330**	**337**	**347**	**357**	**182**	**215**	**257**	**310**	**323**	**343**	**388**
SSDs	**1,293**	**1,371**	**1,449**	**1,503**	**1,558**	**1,594**	**1,652**	**1,662**	**763**	**885**	**1,065**	**1,185**	**1,406**	**1,620**	**1,796**
Notebook	*846*	*867*	*910*	*919*	*938*	*956*	*975*	*985*	*525*	*600*	*700*	*801*	*884*	*965*	*1,035*
Desktop	*892*	*914*	*960*	*970*	*989*	*1,009*	*1,029*	*1,039*	*555*	*632*	*739*	*845*	*935*	*1,017*	*1,091*
Server	*3,822*	*3,899*	*3,996*	*4,076*	*4,158*	*4,241*	*4,347*	*4,434*	*2,348*	*2,676*	*3,215*	*3,640*	*3,959*	*4,305*	*4,683*
Spot & inventories	*617*	*632*	*645*	*658*	*691*	*708*	*722*	*736*	*388*	*438*	*505*	*572*	*639*	*716*	*801*
Tablets	370	351	341	344	344	351	358	373	223	269	324	383	352	357	390
Game - NAND embedded	941	960	970	979	1,009	1,029	1,039	1,049	719	779	841	900	965	1,034	1,108
Autos - NAND embedded	91	95	101	107	112	118	125	133	38	48	65	80	99	122	152
NAND-based products (mn units)															
Smartphones	**275**	**244**	**247**	**259**	**233**	**236**	**247**	**260**	**1,198**	**1,152**	**1,218**	**1,237**	**1,025**	**976**	**1,028**
SSDs	**97**	**94**	**100**	**105**	**90**	**93**	**104**	**113**	**399**	**351**	**378**	**425**	**396**	**401**	**426**
Notebook	*43*	*37*	*37*	*37*	*33*	*35*	*39*	*43*	*193*	*175*	*182*	*197*	*155*	*149*	*165*
Desktop	*15*	*15*	*16*	*17*	*14*	*14*	*16*	*16*	*57*	*52*	*55*	*65*	*63*	*61*	*59*
Server	*16*	*17*	*19*	*22*	*19*	*20*	*23*	*24*	*58*	*53*	*60*	*64*	*74*	*86*	*95*
Spot & inventories	*23*	*24*	*27*	*29*	*24*	*25*	*27*	*30*	*91*	*71*	*80*	*98*	*103*	*106*	*107*
Tablets	34	31	29	31	29	29	28	32	163	138	147	151	125	118	124
Game - NAND embedded	7	7	8	12	7	7	8	12	33	31	32	32	33	34	35
Auto	19	20	21	21	21	21	22	23	68	71	73	77	81	86	92
%QoQ / YoY in NAND-based products															
Smartphones	**-15%**	**-11%**	**1%**	**5%**	**-10%**	**1%**	**5%**	**5%**	**-12%**	**-4%**	**6%**	**2%**	**-17%**	**-5%**	**5%**
SSDs	**-18%**	**-3%**	**6%**	**5%**	**-14%**	**3%**	**12%**	**8%**	**-7%**	**-12%**	**8%**	**12%**	**-7%**	**1%**	**6%**
Notebook	*-18%*	*-12%*	*0%*	*0%*	*-12%*	*6%*	*11%*	*11%*	*-12%*	*-10%*	*4%*	*8%*	*-22%*	*-4%*	*11%*
Desktop	*-13%*	*-1%*	*5%*	*3%*	*-13%*	*0%*	*8%*	*3%*	*10%*	*-9%*	*5%*	*17%*	*-2%*	*-5%*	*-3%*
Server	*-14%*	*7%*	*13%*	*11%*	*-12%*	*4%*	*15%*	*6%*	*8%*	*-9%*	*14%*	*7%*	*15%*	*15%*	*11%*
Spot & inventories	*-25%*	*5%*	*11%*	*10%*	*-19%*	*2%*	*11%*	*10%*	*-15%*	*-22%*	*13%*	*23%*	*5%*	*2%*	*2%*
Tablets	-20%	-10%	-4%	6%	-7%	1%	-4%	13%	-12%	-16%	7%	3%	-17%	-5%	5%
Game - NAND embedded	-35%	-11%	19%	50%	-36%	-10%	25%	40%	-10%	-5%	3%	1%	3%	3%	1%
Auto	-4%	1%	6%	2%	-4%	1%	6%	2%	0%	4%	3%	5%	6%	6%	6%

Source: BofA Global Research estimates, industry data. Chg = change. Equiv = equivalent

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Exhibit 13: Company-specific DRAM sales/ASP/shipment/margin comparison – high margin due to HBM

Bottom-up approach analysis based on top-4 DRAM companies' quarterly earnings results

(8Gb equiv)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
Units (bn)																
Samsung	4.3	4.6	4.7	4.8	4.8	4.9	5.4	5.6	10.9	10.8	11.8	13.4	15.0	18.4	20.7	23.8
SK Hynix	3.1	-	-	-	-	-	-	-	6.8	6.8	7.6	9.0	11.0	-	-	-
Micron	2.5	2.6	2.7	2.9	3.0	3.2	3.4	3.5	5.2	5.2	5.8	7.2	8.6	10.7	13.1	15.4
Nanya	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.6	0.4	0.4	0.4	0.7	0.9	1.0	1.1
Top 4 total	**10.1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**23.5**	**23.3**	**25.7**	**30.1**	**35.3**	**-**	**-**	**-**
ASP (US$)																
Samsung	9.0	13.1	15.8	17.0	17.4	17.4	17.2	16.9	3.9	3.3	1.8	2.9	3.5	13.9	17.2	15.8
SK Hynix	9.4	-	-	-	-	-	-	-	4.1	3.3	2.1	3.6	4.8	-	-	-
Micron	7.5	12.2	14.2	14.6	14.6	14.6	14.2	13.7	4.1	3.8	2.0	2.9	3.9	12.2	14.3	13.0
Nanya	7.9	12.0	14.9	16.0	15.7	15.2	14.9	14.1	5.0	4.5	2.5	2.9	3.5	12.7	15.0	12.5
Top 4 total	**8.7**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**4.0**	**3.4**	**2.0**	**3.1**	**4.0**	**-**	**-**	**-**
OP margin (%)																
Samsung	79%	61%	65%	64%	64%	64%	65%	64%	50%	41%	-4%	30%	40%	66%	64%	59%
SK Hynix	77%	-	-	-	-	-	-	-	39%	31%	0%	47%	61%	-	-	-
Micron	70%	81%	83%	83%	82%	82%	81%	80%	39%	35%	-31%	19%	40%	81%	81%	76%
Nanya	61%	72%	76%	75%	71%	69%	67%	63%	32%	20%	-48%	-31%	7%	72%	68%	55%
Top 4 total	**76%**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**44%**	**36%**	**-10%**	**32%**	**47%**	**-**	**-**	**-**
Sales (US$bn)																
Samsung	39.2	59.7	73.9	81.4	83.2	85.7	92.5	95.2	42.0	35.3	21.8	39.2	53.2	254.2	356.6	377.0
SK Hynix	28.7	-	-	-	-	-	-	-	27.7	22.8	15.7	32.6	52.4	-	-	-
Micron	18.8	31.3	38.9	41.6	43.7	46.3	47.9	48.5	21.6	19.6	11.6	20.6	33.0	130.6	186.5	200.7
Nanya	1.7	2.5	3.2	3.5	3.5	3.6	3.8	3.6	3.1	2.0	1.1	1.3	2.5	11.0	14.5	14.0
Top 4 total	**88.4**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**94.3**	**79.7**	**50.2**	**93.6**	**141.1**	**-**	**-**	**-**
Bit mkt shr																
Samsung	43%	43%	42%	42%	41%	41%	42%	42%	46%	46%	46%	45%	43%	42%	41%	41%
SK Hynix	30%	-	-	-	-	-	-	-	29%	29%	30%	30%	31%	-	-	-
Micron	25%	24%	25%	25%	26%	27%	26%	26%	22%	22%	23%	24%	24%	25%	26%	27%
Nanya	2%	-	-	-	-	-	-	-	3%	2%	2%	1%	2%	-	-	-
Top 4 total	**100%**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**100%**	**100%**	**100%**	**100%**	**100%**	**-**	**-**	**-**
Sales mkt shr																
Samsung	44%	43%	43%	43%	43%	43%	43%	43%	45%	44%	43%	42%	38%	43%	43%	43%
SK Hynix	32%	-	-	-	-	-	-	-	29%	29%	31%	35%	37%	-	-	-
Micron	21%	23%	23%	22%	23%	23%	22%	22%	23%	25%	23%	22%	23%	22%	23%	23%
Nanya	2%	-	-	-	-	-	-	-	3%	3%	2%	1%	2%	-	-	-
Top 4 total	**100%**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**100%**	**100%**	**100%**	**100%**	**100%**	**-**	**-**	**-**
Bit growth QoQ/YoY																
Samsung	3%	5%	3%	2%	0%	3%	9%	5%	27%	-1%	9%	14%	12%	22%	13%	15%
SK Hynix	0%	-	-	-	-	-	-	-	19%	0%	12%	17%	23%	-	-	-
Micron	5%	3%	6%	4%	5%	6%	6%	5%	22%	0%	11%	25%	19%	25%	22%	18%
Nanya	-5%	-3%	2%	1%	3%	4%	8%	1%	0%	-27%	-4%	5%	57%	23%	12%	15%
Top 4 total	**2%**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**23%**	**-1%**	**10%**	**17%**	**17%**	**-**	**-**	**-**
ASP chg QoQ/YoY																
Samsung	91%	45%	20%	8%	2%	0%	-1%	-2%	13%	-15%	-43%	58%	21%	291%	24%	-8%
SK Hynix	65%	-	-	-	-	-	-	-	15%	-18%	-38%	76%	31%	-	-	-
Micron	65%	62%	17%	3%	0%	0%	-3%	-4%	17%	-9%	-47%	43%	35%	217%	17%	-9%
Nanya	71%	51%	25%	7%	-2%	-3%	-2%	-5%	47%	-10%	-44%	13%	24%	259%	18%	-16%
Top 4 total	**76%**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**15%**	**-15%**	**-43%**	**59%**	**28%**	**-**	**-**	**-**

***Upbeat YTD data points not yet fully reflected in 2026 ASP; thus earnings upside seen**

****SK Hynix is based on only historical data; total includes other companies, thus Hynix forecast number cannot be derived**

Source: Companies, BofA Global Research estimates, industry data. Chg = change. Equiv = equivalent

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Exhibit 14: Company-specific NAND sales/ASP/shipment/margin comparison – high margin due to eSSD and 3D NAND

Bottom-up approach analysis based on top-4 NAND companies' quarterly earnings results

(256Gb equiv)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2022	2023	2024	2025E	2026E	2027E	2028E
Sales (US$bn)															
Samsung	15.4	25.3	32.4	34.9	33.2	32.2	33.1	34.6	22.2	14.2	26.1	25.2	108.0	133.1	133.6
Kioxia ex-foundry	5.8	10.4	12.5	12.7	11.7	11.6	12.1	12.1	9.4	5.6	9.5	9.2	41.4	47.4	46.7
Micron	5.0	9.9	11.4	12.5	12.9	12.7	12.2	12.4	7.0	4.3	8.2	9.0	38.9	50.2	50.3
SK Hynix	9.7	-	-	-	-	-	-	-	12.4	8.4	16.0	16.9	-	-	-
Global total	**45.3**	**78.9**	**94.7**	**100.6**	**99.9**	**101.3**	**104.9**	**106.9**	**67.3**	**42.4**	**78.1**	**81.0**	**319.6**	**413.0**	**414.1**
OP margin (%)															
Samsung	60%	62%	64%	63%	62%	61%	59%	57%	13%	-68%	13%	4%	63%	60%	56%
Kioxia ex-foundry	60%	72%	72%	70%	67%	65%	63%	61%	15%	-46%	27%	19%	70%	64%	53%
Micron	58%	78%	78%	79%	78%	77%	75%	74%	0%	-71%	8%	5%	76%	76%	71%
SK Hynix	54%	-	-	-	-	-	-	-	-12%	-76%	14%	8%	-	-	-
ASP (US$)															
Samsung	5.5	8.8	10.1	10.1	9.9	9.6	9.2	8.9	2.9	1.6	2.6	2.5	8.8	9.4	8.3
Kioxia ex-foundry	5.2	9.2	9.5	9.3	9.0	8.6	8.1	7.9	3.4	1.8	2.6	2.2	8.4	8.4	7.2
Micron	2.3	4.3	4.6	4.8	4.7	4.6	4.2	4.1	1.7	0.8	1.3	1.2	4.1	4.4	3.7
SK Hynix	5.5	-	-	-	-	-	-	-	2.4	1.4	2.6	2.5	-	-	-
Global total	**5.0**	**8.3**	**9.4**	**9.5**	**9.3**	**9.0**	**8.6**	**8.3**	**3.0**	**1.6**	**2.6**	**2.4**	**8.1**	**8.8**	**7.6**
Units (bn)															
Samsung	2.8	2.9	3.2	3.4	3.3	3.3	3.6	3.9	7.7	8.9	9.9	10.2	12.3	14.2	16.2
Kioxia ex-foundry	1.1	1.1	1.3	1.4	1.3	1.3	1.5	1.5	2.7	3.0	3.6	4.1	4.9	5.7	6.5
Micron	2.1	2.3	2.5	2.6	2.7	2.8	2.9	3.1	4.1	5.3	6.2	7.8	9.5	11.4	13.7
SK Hynix	1.8	-	-	-	-	-	-	-	5.1	6.1	6.0	6.7	-	-	-
Global total	**9.0**	**9.5**	**10.1**	**10.6**	**10.7**	**11.2**	**12.2**	**12.8**	**22.6**	**26.6**	**29.5**	**33.2**	**39.3**	**47.0**	**54.3**
Bit growth QoQ/YoY															
Samsung	7%	3%	11%	8%	-3%	0%	8%	8%	3%	16%	11%	3%	20%	15%	14%
Kioxia ex-foundry	-10%	2%	16%	4%	-5%	3%	11%	3%	-1%	10%	19%	15%	20%	15%	15%
Micron	2%	6%	9%	5%	4%	2%	4%	5%	0%	27%	18%	26%	22%	20%	19%
SK Hynix	-10%	-	-	-	-	-	-	-	42%	21%	-2%	12%	-	-	-
Global total	**-2%**	**5%**	**6%**	**5%**	**1%**	**5%**	**9%**	**5%**	**6%**	**18%**	**11%**	**13%**	**18%**	**20%**	**16%**
ASP chg QoQ/YoY															
Samsung	88%	60%	15%	0%	-2%	-3%	-5%	-3%	-15%	-45%	66%	-7%	256%	7%	-12%
Kioxia ex-foundry	105%	75%	3%	-2%	-3%	-4%	-6%	-3%	-15%	-46%	43%	-15%	276%	0%	-14%
Micron	79%	87%	6%	4%	-2%	-3%	-8%	-4%	-3%	-52%	61%	-13%	254%	7%	-16%
SK Hynix	75%	-	-	-	-	-	-	-	-18%	-44%	93%	-5%	-	-	-
Global total	**75%**	**65%**	**13%**	**1%**	**-2%**	**-3%**	**-5%**	**-3%**	**-13%**	**-46%**	**65%**	**-8%**	**234%**	**8%**	**-13%**
Sales mkt shr															
Samsung	34%	32%	34%	35%	33%	32%	32%	32%	33%	34%	33%	31%	34%	32%	32%
Kioxia ex-foundry	13%	13%	13%	13%	12%	11%	11%	11%	14%	13%	12%	11%	13%	11%	11%
Micron	11%	13%	12%	12%	13%	13%	12%	12%	10%	10%	11%	11%	12%	12%	12%
SK Hynix	21%	-	-	-	-	-	-	-	18%	20%	20%	21%	-	-	-
Global total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
Bit mkt shr															
Samsung	31%	30%	32%	32%	31%	30%	30%	30%	34%	34%	34%	31%	31%	30%	30%
Kioxia ex-foundry	12%	12%	13%	13%	12%	12%	12%	12%	12%	11%	12%	12%	13%	12%	12%
Micron	24%	24%	25%	25%	25%	25%	24%	24%	18%	20%	21%	23%	24%	24%	25%
SK Hynix	20%	-	-	-	-	-	-	-	23%	23%	20%	20%	-	-	-
Global total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**

***Upbeat YTD data points not yet fully reflected in 2026 ASP; thus earnings upside seen**

****SK Hynix is based on only historical data; total includes other companies, thus Hynix forecast number cannot be derived**

Source: Companies, BofA Global Research estimates, industry data. Chg = change. Equiv = equivalent

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Exhibit 15: Smartphone and PC production cut assumed for 2026 due to memory chip shortage and very expensive BOM – a mild recovery assumed for 2027-28 once memory supply shortage somewhat relieved; separately server shipment growth can be stronger due to AI boom

Product unit assumptions – our underlying assumptions for tech product shipments

	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	CAGR 2025-28E
Units (mn)														
5G smartphone				*19*	*269*	*614*	*678*	*730*	*847*	*888*	*752*	*777*	*850*	-1.4%
Non-5G smartphone	*1,488*	*1,508*	*1,432*	*1,394*	*1,031*	*744*	*520*	*421*	*371*	*349*	*273*	*199*	*178*	-20.0%
Smartphones	**1,488**	**1,508**	**1,432**	**1,413**	**1,300**	**1,358**	**1,198**	**1,152**	**1,218**	**1,237**	**1,025**	**976**	**1,028**	-6.0%
Apple (iPhone)	*215*	*216*	*205*	*192*	*195*	*238*	*233*	*225*	*222*	*243*	*215*	*262*	*269*	3.4%
Samsung	*311*	*315*	*290*	*295*	*254*	*271*	*258*	*225*	*224*	*240*	*223*	*232*	*242*	0.3%
Other OEMs	*962*	*977*	*936*	*926*	*851*	*849*	*707*	*702*	*772*	*754*	*587*	*482*	*517*	-11.8%
Tablet	204	185	173	160	188	184	163	138	147	151	125	118	124	-6.4%
SSD	**130**	**167**	**201**	**248**	**340**	**431**	**399**	**351**	**378**	**425**	**396**	**401**	**426**	0.1%
PC	**256**	**255**	**255**	**261**	**297**	**351**	**294**	**253**	**256**	**275**	**228**	**217**	**228**	-6.0%
Server	10	10	12	12	13	14	15	12	14	17	20	23	25	12.9%
Auto - DRAM embedded	**57**	**61**	**65**	**67**	**63**	**69**	**69**	**72**	**75**	**79**	**85**	**90**	**96**	6.7%
TV (LCD, OLED)	222	215	221	223	225	214	203	199	201	203	206	209	213	1.6%
Game - DRAM embedded	**44**	**50**	**52**	**51**	**42**	**48**	**43**	**41**	**42**	**42**	**43**	**45**	**45**	2.1%
Growth YoY														
5G smartphone						*128.1%*	*10.3%*	*7.8%*	*16.0%*	*4.9%*	*-15.3%*	*3.3%*	*9.4%*	
Non-5G smartphone	*3.3%*	*1.3%*	*-5.1%*	*-2.6%*	*-26.1%*	*-27.8%*	*-30.1%*	*-19.0%*	*-12.0%*	*-6.0%*	*-21.8%*	*-27.0%*	*-10.4%*	
Smartphones	**3.3%**	**1.3%**	**-5.1%**	**-1.3%**	**-8.0%**	**4.5%**	**-11.8%**	**-3.9%**	**5.7%**	**1.5%**	**-17.1%**	**-4.8%**	**5.4%**	
Apple (iPhone)	*-7.0%*	*0.2%*	*-4.8%*	*-6.5%*	*1.3%*	*22.2%*	*-2.1%*	*-3.4%*	*-1.3%*	*9.5%*	*-11.5%*	*21.9%*	*2.7%*	
Samsung	*-3.0%*	*1.5%*	*-8.1%*	*1.8%*	*-14.0%*	*6.8%*	*-4.8%*	*-12.8%*	*-0.6%*	*7.3%*	*-7.1%*	*3.9%*	*4.5%*	
Other OEMs	*8.3%*	*1.5%*	*-4.1%*	*-1.1%*	*-8.0%*	*-0.3%*	*-16.8%*	*-0.7%*	*10.0%*	*-2.4%*	*-22.1%*	*-17.8%*	*7.3%*	
Tablet	-9.1%	-9.0%	-6.3%	-7.5%	17.2%	-2.0%	-11.7%	-15.5%	6.6%	3.1%	-17.4%	-5.3%	4.7%	
SSD	**36.2%**	**28.4%**	**20.0%**	**23.7%**	**36.9%**	**26.7%**	**-7.3%**	**-12.0%**	**7.6%**	**12.4%**	**-6.8%**	**1.1%**	**6.2%**	
PC	**-5.9%**	**-0.3%**	**-0.2%**	**2.6%**	**13.6%**	**4.0%**	**-16.2%**	**-13.8%**	**1.0%**	**7.6%**	**-17.0%**	**-4.9%**	**5.1%**	
Server	**-1.6%**	**7.2%**	**15.7%**	**-0.4%**	**7.3%**	**6.9%**	**9.7%**	**-17.5%**	**18.0%**	**19.8%**	**16.9%**	**11.7%**	**10.3%**	
Auto - DRAM embedded	**13.1%**	**7.1%**	**5.3%**	**3.5%**	**-5.2%**	**8.6%**	**0.3%**	**4.7%**	**3.4%**	**5.9%**	**6.8%**	**6.7%**	**6.7%**	
TV (LCD, OLED)	-1.2%	-3.2%	2.9%	0.7%	1.0%	-5.2%	-4.8%	-2.1%	1.0%	1.1%	1.2%	1.8%	1.8%	
Game - DRAM embedded	**6.9%**	**12.8%**	**3.0%**	**-1.3%**	**-16.8%**	**12.5%**	**-9.7%**	**-4.9%**	**2.5%**	**1.0%**	**2.4%**	**3.0%**	**0.8%**	

Note: Sell-in based thus actual sell-through can be higher/lower than our assumptions; we still assume minimal impact of Huawei given other OEMs' market share increase, etc; memory content increase (auto, server, SSD) likely stronger than unit growth; smartphone shipment estimate is slightly lowered due to Samsung Electronics' weak shipment for mid to low end phones vs upbeat AI smartphone GS24

Source: SA, IDC, IHS, BofA Global Research estimates

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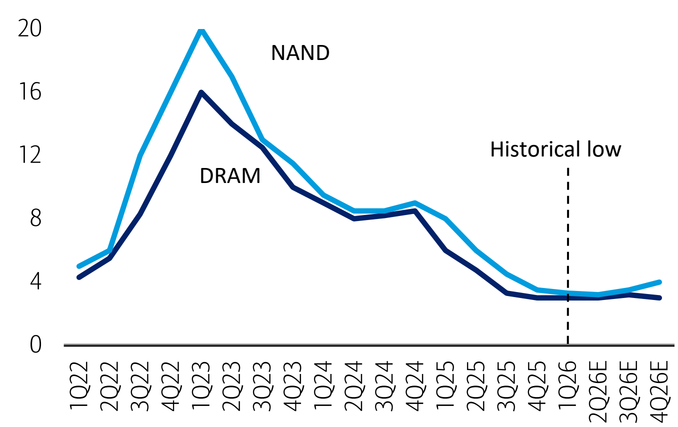
Exhibit 16: Only a few weeks for both DRAM and NAND as of April 2026, much lower than normal (1-2 months); 2H26 inventories likely up following fab utilization increase but still tight (3-4 weeks likely)

Memory chipmakers' inventories - finished chips in weeks



Source: BofA Global Research estimates

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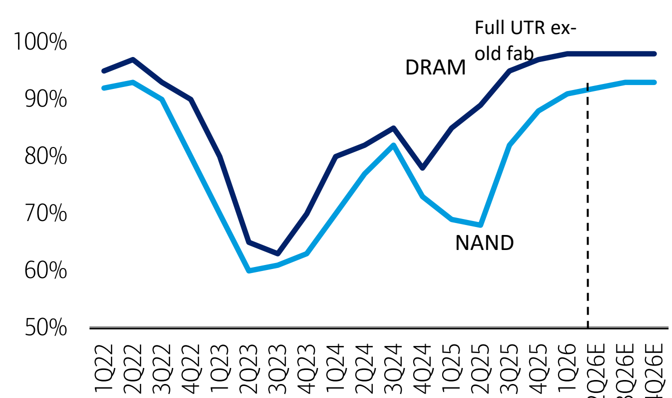
Exhibit 17: Both DRAM and NAND fabs fully utilized ex old idle capacity as of April 2026

Memory chipmakers' fab utilization



Source: BofA Global Research estimates

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Exhibit 18: Mostly server (including HBM) and smartphone-driven sales seen
Implied DRAM sales mix based on bit shipments x ASP by application

	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
DRAM sales mix													
Servers	25%	26%	35%	24%	30%	31%	41%	33%	48%	55%	58%	61%	65%
PCs	10%	14%	14%	13%	14%	15%	14%	11%	11%	12%	6%	5%	5%
Smartphones	26%	31%	29%	45%	38%	36%	34%	33%	34%	30%	21%	18%	17%
Spot, others, adjustments*	39%	29%	22%	18%	18%	18%	12%	22%	7%	2%	14%	16%	13%
Demand mix total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
DRAM sales, US$bn													
Servers	10.2	18.9	34.4	14.9	19.4	28.5	31.6	15.8	42.3	73.4	325.2	485.3	566.1
PCs	4.1	10.0	13.9	8.4	8.9	14.2	10.6	5.3	9.6	16.6	36.2	40.5	39.9
Smartphones	10.6	22.9	29.0	27.9	25.0	32.9	26.1	15.7	29.6	40.6	117.4	143.8	150.1
Spot, others, adjustments*	16.3	21.0	22.0	11.3	12.0	16.6	9.4	10.5	6.5	3.2	78.4	129.5	114.9
Total sales US$bn	**41.2**	**72.8**	**99.3**	**62.5**	**65.4**	**92.1**	**77.7**	**47.3**	**87.9**	**133.8**	**557.2**	**799.1**	**870.9**
DRAM ASP, US$/8Gb													
Servers	5.9	7.7	9.9	3.3	3.1	3.6	3.2	1.8	3.2	4.0	13.7	17.2	16.4
PCs	2.6	5.8	7.2	3.7	3.0	3.6	3.0	1.5	2.4	3.5	10.2	12.6	10.8
Smartphones	4.1	6.6	6.9	5.1	4.1	4.4	3.5	2.0	2.9	3.3	11.4	14.4	13.0
Spot, others, adjustments*	4.5	5.7	6.6	3.0	2.9	3.4	2.7	1.7	2.0	4.6	12.4	13.3	10.7
DRAM ind avg price	**4.3**	**6.4**	**7.7**	**3.9**	**3.3**	**3.8**	**3.2**	**1.8**	**2.9**	**3.7**	**12.7**	**15.6**	**14.4**
ASP change YoY													
Servers	-20%	31%	29%	-66%	-7%	17%	-11%	-46%	82%	24%	245%	25%	-4%
PCs	-48%	122%	26%	-49%	-18%	21%	-19%	-48%	58%	45%	188%	24%	-15%
Smartphones	-42%	60%	5%	-26%	-20%	8%	-20%	-45%	48%	16%	242%	26%	-10%
Spot, others, adjustments*	-35%	25%	17%	-54%	-6%	19%	-21%	-37%	18%	128%	172%	8%	-20%
DRAM ind avg price	**-34%**	**48%**	**20%**	**-49%**	**-15%**	**14%**	**-16%**	**-45%**	**62%**	**29%**	**242%**	**23%**	**-8%**

*Implied amount after server/PC/smartphone demand, thus negative figures possible due to channel inventories (not consumed)
Source: Company data, BofA Global Research estimates, industry data

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Exhibit 19: Mostly for SSD (including enterprise solutions) and smartphone applications
Implied NAND sales mix based on bit shipments x ASP by application

	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
NAND sales mix													
SSD	30%	33%	34%	36%	45%	47%	48%	40%	49%	52%	60%	61%	64%
Smartphone	36%	38%	33%	34%	33%	34%	33%	31%	35%	33%	25%	23%	23%
Spot, others, adjustments*	34%	29%	33%	30%	22%	19%	18%	29%	16%	14%	15%	17%	14%
Demand mix total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
NAND sales, US$bn													
SSD	11.1	19.8	22.9	16.7	26.5	34.3	32.6	16.9	38.6	42.2	191.3	251.3	263.8
Smartphone	13.7	22.3	22.0	15.5	19.2	24.9	22.4	13.1	27.2	27.1	80.9	93.0	94.0
Spot, others, adjustments*	12.8	17.2	22.6	14.1	12.7	14.1	12.3	12.4	12.3	11.7	47.4	68.8	56.2
Total sales US$bn	**37.6**	**59.2**	**67.4**	**46.3**	**58.5**	**73.3**	**67.3**	**42.4**	**78.1**	**81.0**	**319.6**	**413.0**	**414.1**
NAND ASP, US$/8Gb													
SSD	8.5	10.6	8.8	4.5	4.4	4.0	3.4	1.7	3.1	2.7	11.0	12.4	11.0
Smartphone	8.7	10.5	8.6	4.4	4.3	4.1	3.3	1.7	2.8	2.3	7.8	8.9	7.5
Spot, others, adjustments*	7.7	9.2	6.9	3.0	2.6	2.1	2.0	1.4	1.7	2.1	4.1	4.2	3.1
NAND ind avg price	**8.3**	**10.1**	**8.0**	**3.9**	**3.8**	**3.4**	**3.0**	**1.6**	**2.6**	**2.4**	**8.1**	**8.8**	**7.6**
ASP change YoY													
SSD	-29%	25%	-17%	-49%	-1%	-10%	-13%	-49%	76%	-13%	310%	13%	-11%
Smartphone	-30%	21%	-18%	-48%	-2%	-4%	-21%	-48%	64%	-18%	245%	14%	-15%
Spot, others, adjustments*	-27%	19%	-24%	-57%	-14%	-17%	-8%	-31%	26%	23%	95%	3%	-26%
NAND ind avg price	**-28%**	**22%**	**-21%**	**-52%**	**-2%**	**-9%**	**-13%**	**-46%**	**65%**	**-8%**	**234%**	**8%**	**-13%**

*Implied amount after SSD/smartphone demand, thus negative figures possible due to channel inventories (not consumed)
Source: Company data, BofA Global Research estimates, industry data

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Record high capex includes heavy infrastructure spending

Exhibit 20: Big-3 DRAM makers' capex to grow in 2026, but it includes lots of non-WFE spending (shell fab, utilities, TCB/HB); overall capex growth rate likely to decelerate through 2027-28 vs 40%+ CAGR in 2023-25

DRAM capex trend

US$bn	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Samsung	12.5	8.9	8.2	11.6	12.2	13.9	17.0	19.9	29.3	33.7	37.1
SK Hynix	8.1	7.4	5.6	7.3	8.9	3.8	9.0	16.1	-	-	-
Micron	4.6	4.3	5.3	5.6	5.7	3.9	6.3	13.1	27.9	37.4	37.2
Nanya	0.7	0.2	0.3	0.4	0.7	0.4	0.5	0.4	1.8	2.3	2.3
CXMT		0.7	1.5	1.5	1.7	2.6	4.0	3.5	4.0	4.4	4.5
DRAM capex total	**27.1**	**21.9**	**21.4**	**27.1**	**31.0**	**25.2**	**37.7**	**53.9**	**89.1**	**107.9**	**112.3**
Capex change YoY	**66%**	**-19%**	**-2%**	**27%**	**14%**	**-19%**	**49%**	**43%**	**65%**	**21%**	**4%**
DRAM capex to sales	**27%**	**35%**	**33%**	**29%**	**40%**	**53%**	**43%**	**40%**	**16%**	**13%**	**13%**
DRAM capex share											
Samsung	46%	41%	38%	43%	39%	55%	45%	37%	33%	31%	33%
SK Hynix	30%	34%	26%	27%	29%	15%	24%	30%	-	-	-
Micron	17%	19%	25%	21%	19%	16%	17%	24%	31%	35%	33%
Nanya	2%	1%	1%	1%	2%	2%	1%	1%	2%	2%	2%
CXMT		3%	7%	6%	5%	10%	11%	6%	4%	4%	4%
Total	**99%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
DRAM capex YoY											
Samsung	72%	-28%	-8%	42%	5%	14%	22%	17%	47%	15%	10%
SK Hynix	71%	-10%	-23%	30%	22%	-58%	138%	79%	-	-	-
Micron	72%	-8%	24%	6%	3%	-32%	62%	106%	114%	34%	0%
Nanya	-30%	-74%	62%	40%	72%	-39%	27%	-17%	309%	27%	0%
CXMT		75%	114%	0%	13%	50%	57%	-13%	14%	10%	2%
Total	**66%**	**-19%**	**-2%**	**27%**	**14%**	**-19%**	**49%**	**43%**	**65%**	**21%**	**4%**

***SK Hynix is based on only historical data; total includes other companies, thus Hynix forecast number cannot be derived**

Source: Companies' reports, BofA Global Research estimates

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Exhibit 21: NAND capex also grows in 2026, but mainly due to low base in 2025

NAND capex trend

US$bn	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Samsung	7.3	4.9	9.2	10.9	10.3	10.1	8.1	4.6	8.3	10.7	11.6
Kioxia & JVs	5.0	3.8	4.5	5.7	6.5	2.7	4.5	5.2	7.0	7.5	7.0
Micron	4.7	4.4	3.4	4.6	5.2	2.7	3.1	4.6	7.4	7.7	7.9
SK Hynix	6.3	4.4	2.7	3.4	4.1	1.6	2.3	2.5	-	-	-
Intel China	2.0	1.6	0.8	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
YMTC	0.3	0.7	1.6	1.5	2.5	2.3	2.0	2.0	3.0	3.5	4.0
NAND capex total	**26.0**	**20.1**	**22.5**	**27.9**	**29.0**	**19.7**	**20.4**	**19.3**	**30.1**	**34.8**	**36.3**
NAND capex chg YoY	**7%**	**-23%**	**12%**	**24%**	**4%**	**-32%**	**4%**	**-5%**	**56%**	**16%**	**4%**
NAND capex to sales	**38%**	**43%**	**38%**	**38%**	**43%**	**46%**	**26%**	**24%**	**9%**	**8%**	**9%**
NAND capex share											
Samsung	28%	25%	41%	39%	36%	51%	39%	24%	27%	31%	32%
Kioxia & JVs	19%	19%	20%	20%	22%	14%	22%	27%	23%	22%	19%
Micron	18%	22%	15%	17%	18%	14%	15%	24%	24%	22%	22%
SK Hynix	24%	22%	12%	12%	14%	8%	11%	13%	-	-	-
Intel China	8%	8%	4%	5%	n/a	n/a	n/a	n/a	n/a	n/a	n/a
YMTC	1%	3%	7%	5%	9%	11%	10%	10%	10%	10%	11%
NAND capex total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
NAND capex growth (YoY)											
Samsung	-36%	-33%	87%	19%	-5%	-2%	-20%	-42%	78%	29%	9%
Kioxia & JVs	4%	-24%	18%	27%	14%	-58%	67%	16%	35%	7%	-7%
Micron	100%	-6%	-22%	35%	11%	-47%	13%	49%	60%	5%	2%
SK Hynix	97%	-29%	-38%	26%	19%	-60%	42%	5%	-	-	-
Intel China	0%	-23%	-48%	88%	na	na	na	na	na	na	na
YMTC	200%	133%	129%	-6%	67%	-10%	-11%	0%	50%	17%	14%
NAND capex total	**7%**	**-23%**	**12%**	**24%**	**4%**	**-32%**	**4%**	**-5%**	**56%**	**16%**	**4%**

***SK Hynix is based on only historical data; total includes other companies, thus Hynix forecast number cannot be derived**

Source: Companies' reports, BofA Global Research estimates

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Exhibit 22: Ex-conventional DRAM equipment capex spend likely up strongly due to HBM, but conventional DRAM and NAND spend broadly disciplined; non-equipment spend for buildings and utilities remain high

SPE vs non-SPE capex spend

US$bn	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
SPE capex										
DRAM	12.1	12.7	16.2	17.3	14.1	23.8	34.8	59.2	71.6	75.2
NAND	12.1	14.3	18.2	18.7	11.9	12.9	13.5	21.5	25.2	26.1
Total	**24.3**	**26.9**	**34.4**	**35.9**	**26.0**	**36.7**	**48.3**	**80.7**	**96.8**	**101.4**
SPE capex YoY										
DRAM	-22%	4%	28%	7%	-18%	69%	46%	70%	21%	5%
NAND	-24%	17%	28%	3%	-36%	8%	5%	59%	17%	4%
Total	**-23%**	**11%**	**28%**	**5%**	**-28%**	**41%**	**32%**	**67%**	**20%**	**5%**
Non-SPE to capex										
DRAM	45%	41%	40%	44%	44%	37%	35%	34%	34%	33%
NAND	39%	37%	35%	36%	40%	37%	30%	29%	28%	28%
Total	**42%**	**39%**	**38%**	**40%**	**42%**	**37%**	**34%**	**32%**	**32%**	**32%**
Total capex, US$bn										
DRAM	21.9	21.4	27.1	31.0	25.2	37.7	53.9	89.1	107.9	112.3
NAND	20.1	22.5	27.9	29.0	19.7	20.4	19.3	30.1	34.8	36.3
Total	**42.0**	**43.9**	**55.0**	**60.0**	**45.0**	**58.2**	**73.2**	**119.2**	**142.7**	**148.6**

Source: Company data, BofA Global Research estimates, industry data

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Exhibit 23: DRAM capex increase likely to remain strong even in 2026 driven by big-3 chipmakers; NAND capex growth also notable (due to low base)

SPE capex spend by memory chipmakers

US$bn	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
DRAM SPE										
Samsung	5.5	4.8	6.9	7.0	7.8	9.8	12.3	18.5	21.6	24.1
SK Hynix	2.9	2.5	3.4	4.3	1.6	6.1	9.9	-	-	-
Micron	2.9	3.5	3.7	3.8	2.1	3.8	9.0	18.1	23.2	23.4
Nanya	0.1	0.2	0.4	0.5	0.3	0.4	0.3	1.2	1.5	1.5
CXMT	0.3	1.2	1.3	1.2	2.0	3.2	2.8	3.2	3.5	3.6
DRAM SPE total	**12.1**	**12.7**	**16.2**	**17.3**	**14.1**	**23.8**	**34.8**	**59.2**	**71.6**	**75.2**
YoY										
Samsung	-25%	-14%	44%	1%	12%	27%	25%	50%	17%	12%
SK Hynix	-21%	-14%	32%	27%	-63%	284%	63%	-	-	-
Micron	-14%	24%	6%	3%	-45%	80%	137%	101%	28%	1%
DRAM SPE total	**-22%**	**4%**	**28%**	**7%**	**-18%**	**69%**	**46%**	**70%**	**21%**	**5%**
NAND SPE										
Samsung	3.4	5.7	6.7	6.6	6.2	5.2	3.2	6.2	8.0	8.7
Kioxia & JVs	2.1	2.7	3.4	4.1	1.4	2.7	3.7	5.0	5.3	4.8
Micron	3.0	2.3	3.1	3.5	1.4	1.9	3.0	5.2	5.4	5.5
SK Hynix	2.3	1.8	2.4	2.5	1.0	1.5	1.9	-	-	-
YMTC	0.0	1.0	1.2	1.8	1.8	1.3	1.4	2.0	2.5	2.8
NAND SPE total	**12.1**	**14.3**	**18.2**	**18.7**	**11.9**	**12.9**	**13.5**	**21.5**	**25.2**	**26.1**
YoY										
Samsung	-26%	68%	19%	-2%	-6%	-16%	-40%	96%	29%	9%
Micron	-3%	-22%	35%	11%	-59%	33%	62%	72%	5%	1%
SK Hynix	-38%	-24%	36%	2%	-61%	57%	27%	-	-	-
YMTC			25%	46%	3%	-28%	8%	39%	26%	14%
NAND SPE total	**-24%**	**17%**	**28%**	**3%**	**-36%**	**8%**	**5%**	**59%**	**17%**	**4%**

***SK Hynix is based on only historical data; total includes other companies, thus Hynix forecast number cannot be derived**

Source: Company data, BofA Global Research estimates, industry data

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2026/27E HBM TAM expansion to US$77bn/135bn

Exhibit 24: We maintain our bullish view on HBM. We assume strong growth in 2026 sales (US$77bn; +122% YoY) driven by solid ASP growth (+27% YoY) and robust bit growth (+76%). Industry average OPM near 50%; no downturn in 2027 given volume growth and cost reduction fully offset price cut.
Global HBM market – record high revenue continues into 2027E (US$135bn)

	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	CAGR 2025-28E
HBM earnings										
Sales, US$bn	**1.6**	**4.2**	**17.4**	**34.5**	**76.8**	**134.6**	**176.3**	**204.6**	**246.3**	**72%**
Shipment units, GB bn	0.1	0.5	1.6	3.1	5.4	7.7	10.1	12.9	16.2	49%
ASP, US$/GB	**16.7**	**8.8**	**11.0**	**11.3**	**14.3**	**17.4**	**17.5**	**15.9**	**15.2**	**16%**
Price premium	**430%**	**436%**	**355%**	**274%**	**14%**	**14%**	**27%**	**7%**	**-2%**	
OP, US$bn	0.4	1.8	8.3	14.6	37.6	72.4	98.2	111.3	135.7	89%
OP margin	**25%**	**42%**	**48%**	**42%**	**49%**	**54%**	**56%**	**54%**	**55%**	
% YoY										
Sales, US$bn	55%	155%	316%	98%	122%	75%	31%	16%	20%	
Shipment units, GB bn	95%	383%	233%	94%	76%	44%	30%	27%	26%	
ASP, US$/GB	n/a	n/a	25%	2%	27%	22%	0%	-9%	-4%	
HBM shipment mix										
HBM5+	0%	0%	0%	0%	0%	0%	23%	55%	79%	
HBM4e	0%	0%	0%	0%	0%	19%	51%	36%	19%	
HBM4	0%	0%	0%	0%	32%	64%	24%	9%	3%	
HBM3e	0%	0%	58%	83%	60%	16%	3%	1%	0%	
HBM3	10%	60%	27%	14%	7%	1%	0%	0%	0%	
HBM2e & below	90%	40%	15%	3%	1%	0%	0%	0%	0%	
HBM mix total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	
% to DRAM shipment	**0.4%**	**1.8%**	**5.2%**	**8.5%**	**12.2%**	**15.1%**	**16.7%**	**18.8%**	**21.1%**	
% to DRAM sales	**2.1%**	**8.8%**	**19.8%**	**25.8%**	**13.8%**	**16.8%**	**20.2%**	**19.8%**	**20.9%**	
Wafer capacity, k wpm	**68**	**87**	**204**	**360**	**476**	**624**	**741**	**815**	**897**	**27%**
% to DRAM wafer capacity	**4.4%**	**5.4%**	**11.8%**	**19.3%**	**23.0%**	**27.8%**	**31.1%**	**32.6%**	**34.5%**	
Main node	1z	1z	1a, 1b	1a, 1b	1b, 1c	1c	1c, 1d	1d	1d, 1e	
Implied demand, GB bn units	0.1	0.5	1.8	3.3	6.4	8.7	10.2	12.1	15.8	46%
AI+HBM server units, mn	**0.3**	**1.0**	**1.9**	**2.8**	**4.5**	**5.9**	**6.3**	**6.8**	**7.5**	**32%**
AI+HBM server shipment YoY	42%	289%	88%	50%	62%	30%	8%	7%	11%	
GB HBM per AI+HBM server	**340**	**530**	**954**	**1,171**	**1,413**	**1,479**	**1,612**	**1,789**	**2,093**	**11%**
GB content growth QoQ/YoY	13%	56%	80%	23%	21%	5%	9%	11%	17%	
AI+HBM to total server units*	1.7%	8.1%	12.9%	16.1%	22.3%	26.0%	25.4%	25.4%	26.4%	
Global server units, mn	14.9	12.3	14.5	17.3	20.2	22.6	24.9	26.7	28.6	13%
Global server units YoY	10%	-17%	18%	20%	17%	12%	10%	7%	7%	
Excess supply, GB bn units	0.01	-0.05	-0.20	-0.20	-1.01	-0.95	-0.15	0.70	0.41	
HBM sufficiency (supply to demand)	13%	-10%	-11%	-6%	-16%	-11%	-1%	6%	3%	

Source: Companies' reports, BofA Global Research estimates

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Exhibit 25: Our bottom-up approach based on Big-3 companies' track record and long-term plans

Global HBM market share by company

	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	CAGR 2025-28E
HBM units, GB bn										
SK Hynix	0.06	0.27	0.87	1.78	-	-	-	-	-	-
Samsung	0.03	0.20	0.61	0.66	1.82	2.80	3.61	4.51	5.50	76%
Others	0.00	0.01	0.10	0.62	-	-	-	-	-	-
Global HBM total GB bn units	**0.10**	**0.47**	**1.58**	**3.06**	**5.38**	**7.74**	**10.08**	**12.85**	**16.19**	**49%**
% of global DRAM volume	**0.4%**	**1.8%**	**5.2%**	**8.5%**	**12.2%**	**15.1%**	**16.7%**	**18.8%**	**21.1%**	
HBM wafer capa, k wpm - yr avg										
SK Hynix	35	37	100	153	-	-	-	-	-	-
Samsung	30	45	78	133	178	246	298	312	322	31%
Others	3	5	27	75	-	-	-	-	-	-
Global HBM wafer capa total	**68**	**87**	**204**	**360**	**476**	**624**	**741**	**815**	**897**	**27%**
% of global DRAM capacity	**4.4%**	**5.4%**	**11.8%**	**19.3%**	**23.0%**	**27.8%**	**31.1%**	**32.6%**	**34.5%**	
Main node	1z	1z	1a, 1b	1a, 1b	1b, 1c	1c	1c, 1d	1d	1d, 1e	
HBM ASP, US$/GB										
SK Hynix	16.2	8.5	11.1	11.8	-	-	-	-	-	-
Samsung	17.8	9.3	11.0	9.8	14.8	17.3	17.4	16.0	15.2	21%
Global average - HBM ASP	**16.7**	**8.8**	**11.0**	**11.3**	**14.3**	**17.4**	**17.5**	**15.9**	**15.2**	**16%**
HBM price premium	**430%**	**436%**	**355%**	**274%**	**14%**	**14%**	**27%**	**7%**	**-2%**	
HBM sales, US$bn										
SK Hynix	1.1	2.3	9.6	20.9	-	-	-	-	-	-
Samsung	0.6	1.8	6.7	6.5	26.9	48.5	62.7	72.1	83.6	113%
Others	0.0	0.1	1.1	7.1	-	-	-	-	-	-
Global HBM sales total	**1.6**	**4.2**	**17.4**	**34.5**	**76.8**	**134.6**	**176.3**	**204.6**	**246.3**	**72%**
% of global DRAM sales	**2.1%**	**8.8%**	**19.8%**	**25.8%**	**13.8%**	**16.8%**	**20.2%**	**19.8%**	**20.9%**	
Ex-HBM global DRAM										
ex-HBM wafer capa, kwpm	**1,493**	**1,541**	**1,530**	**1,507**	**1,590**	**1,618**	**1,638**	**1,683**	**1,701**	**3%**
ex-HBM GB bn units	**24.1**	**26.2**	**29.0**	**33.0**	**38.5**	**43.5**	**50.4**	**55.5**	**60.4**	**15%**
ex-HBM ASP, US$/GB	**3.2**	**1.6**	**2.4**	**3.0**	**12.5**	**15.3**	**13.8**	**14.9**	**15.5**	**66%**
ex-HBM sales, US$bn	**76.0**	**43.1**	**70.5**	**99.3**	**480.4**	**664.5**	**694.7**	**828.7**	**934.2**	**91%**
HBM OPM										
SK Hynix	25%	47%	63%	64%	-	-	-	-	-	-
Samsung	27%	38%	33%	20%	53%	58%	62%	62%	63%	
HBM OPM weighted avg	**26%**	**42%**	**49%**	**50%**	**59%**	**65%**	**67%**	**66%**	**67%**	
HBM bit market share										
SK Hynix	66%	57%	55%	58%	-	-	-	-	-	-
Samsung	32%	42%	39%	22%	34%	36%	36%	35%	34%	
HBM sales market share										
SK Hynix	64%	54%	55%	61%	-	-	-	-	-	-
Samsung	34%	44%	39%	19%	35%	36%	36%	35%	34%	
HBM wafer capa mkt share										
SK Hynix	51%	43%	49%	42%	-	-	-	-	-	-
Samsung	44%	52%	38%	37%	37%	39%	40%	38%	36%	
HBM wafer capa mkt share total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	

***SK Hynix is based on only historical data; total includes other companies, thus Hynix forecast number cannot be derived**

Source: Companies' reports, BofA Global Research estimates

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Exhibit 26: Samsung's upside could come from more normalized HBM price (2026-28) vs 2025 bottom, focused on NVIDIA's Tier 2 customers (ex-US data centers) and Big Tech companies (Amazon, Google, Broadcom, AMD, etc.)

SEC – HBM earnings outlook

	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
HBM																
Sales, US$bn	**2.82**	**4.45**	**7.33**	**12.34**	**11.17**	**10.93**	**12.50**	**13.90**	**1.84**	**6.75**	**6.46**	**26.94**	**48.50**	**62.73**	**72.14**	**83.61**
% to DRAM sales	7.2%	7.4%	9.9%	15.2%	13.4%	12.8%	13.5%	14.6%	8.4%	17.2%	12.1%	10.6%	13.6%	16.6%	18.3%	21.4%
Shipment, GB bn	**0.24**	**0.30**	**0.48**	**0.80**	**0.68**	**0.64**	**0.71**	**0.77**	**0.20**	**0.61**	**0.66**	**1.82**	**2.80**	**3.61**	**4.51**	**5.50**
ASP, US$/GB	**11.63**	**14.77**	**15.21**	**15.52**	**16.52**	**17.02**	**17.53**	**18.06**	**9.31**	**11.00**	**9.76**	**14.79**	**17.31**	**17.38**	**15.99**	**15.19**
Price premium	31%	14%	-4%	-10%	-6%	-3%	2%	8%	440%	335%	200%	8%	1%	12%	0%	2%
Cost, US$/GB	6.7	6.5	6.9	7.3	7.4	7.4	7.1	7.0	5.8	7.4	7.8	7.0	7.2	6.5	6.1	5.7
OP, US$bn	**1.20**	**2.48**	**4.03**	**6.50**	**6.16**	**6.17**	**7.42**	**8.48**	**0.70**	**2.22**	**1.29**	**14.21**	**28.23**	**39.10**	**44.67**	**52.45**
OP margin	**43%**	**56%**	**55%**	**53%**	**55%**	**56%**	**59%**	**61%**	**38%**	**33%**	**20%**	**53%**	**58%**	**62%**	**62%**	**63%**
Wafer capa, kwpm	**160**	**170**	**180**	**200**	**220**	**240**	**250**	**275**	**45**	**78**	**133**	**178**	**246**	**298**	**312**	**322**
QoQ/YoY																
Sales	27%	57%	65%	68%	-9%	-2%	14%	11%		267%	-4%	317%	80%	29%	15%	16%
Shipment	5%	24%	60%	65%	-15%	-5%	11%	8%		210%	8%	175%	54%	29%	25%	22%
ASP	21%	27%	3%	2%	7%	3%	3%	3%		18%	-11%	52%	17%	0%	-8%	-5%
Cost	13%	-2%	5%	7%	1%	0%	-4%	-1%		28%	6%	-10%	4%	-10%	-7%	-7%
OP	41%	106%	62%	62%	-5%	0%	20%	14%		217%	-42%	998%	99%	39%	14%	17%
Wafer capa	7%	6%	6%	11%	10%	9%	4%	10%		72%	71%	34%	39%	21%	5%	3%
HBM shipment mix																
HBM5 & above	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	39.4%	60.0%	80.0%
HBM4e	0.0%	0.0%	0.0%	0.0%	1.0%	5.0%	15.0%	30.0%	0.0%	0.0%	0.0%	0.0%	13.5%	49.3%	34.0%	18.0%
HBM4	**1.0%**	**20.0%**	**30.0%**	**42.0%**	**60.0%**	**70.0%**	**65.0%**	**60.0%**	**0.0%**	**0.0%**	**0.0%**	**29.7%**	**63.6%**	**11.1%**	**6.0%**	**2.0%**
HBM3e	95.0%	78.0%	69.0%	58.0%	39.0%	25.0%	20.0%	9.9%	0.0%	19.8%	77.3%	69.1%	23.0%	0.2%	0.0%	0.0%
HBM3	3.0%	2.0%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%	6.3%	58.1%	18.4%	1.0%	0.0%	0.0%	0.0%	0.0%
HBM2e & below	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	93.7%	22.1%	4.4%	0.1%	0.0%	0.0%	0.0%	0.0%
Mix total	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**
Hi-mix																
20-hi & above	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	39.4%	60.0%	80.0%
16-hi	0.0%	0.0%	0.0%	0.3%	2.0%	5.0%	15.0%	30.0%	0.0%	0.0%	0.0%	0.1%	13.7%	49.3%	39.0%	20.0%
12-hi	51.0%	75.0%	90.0%	92.7%	94.0%	93.0%	84.0%	69.9%	0.0%	0.4%	17.2%	83.5%	84.6%	11.2%	1.0%	0.0%
8-hi	49.0%	25.0%	10.0%	7.0%	4.0%	2.0%	1.0%	0.0%	100.0%	99.6%	82.8%	16.4%	1.7%	0.0%	0.0%	0.0%
Hi-mix total	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**
% HBM to DRAM																
Sales	7%	7%	10%	15%	13%	13%	14%	15%	8%	17%	12%	11%	14%	17%	18%	21%
Shipment	6%	7%	10%	17%	14%	13%	13%	14%	2%	5%	4%	10%	14%	15%	17%	19%
OP	4%	7%	8%	12%	12%	11%	12%	14%	-72%	19%	6%	8%	12%	18%	38%	45%
Wafer capa	24%	25%	26%	28%	30%	32%	33%	37%	6%	11%	20%	26%	33%	38%	39%	40%
QoQ/YoY	2.82	4.45	7.33	12.34	11.17	10.93	12.50	13.90	1.84	6.75	6.46	26.94	48.50	62.73	72.14	83.61
Sales	7.2%	7.4%	9.9%	15.2%	13.4%	12.8%	13.5%	14.6%	8.4%	17.2%	12.1%	10.6%	13.6%	16.6%	18.3%	21.4%
Shipment	0.24	0.30	0.48	0.80	0.68	0.64	0.71	0.77	0.20	0.61	0.66	1.82	2.80	3.61	4.51	5.50
ASP	11.63	14.77	15.21	15.52	16.52	17.02	17.53	18.06	9.31	11.00	9.76	14.79	17.31	17.38	15.99	15.19
Cost	31%	14%	-4%	-10%	-6%	-3%	2%	8%	440%	335%	200%	8%	1%	12%	0%	2%
OP	6.7	6.5	6.9	7.3	7.4	7.4	7.1	7.0	5.8	7.4	7.8	7.0	7.2	6.5	6.1	5.7
Wafer capa	1.20	2.48	4.03	6.50	6.16	6.17	7.42	8.48	0.70	2.22	1.29	14.21	28.23	39.10	44.67	52.45

Source: Company reports, BofA Global Research estimates

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GPU spec tables

Exhibit 27: H200/B200/B300 will lead to 2025 HBM demand growth and later by Rubin/Rubin Ultra in 2026/27

NVIDIA GPUs that use HBM memory

Company	Name	Type	AI function	Launch date	DRAM spec	Remarks
Mainstream	Feynman	GPU	AI training	2028	HBM5 (likely)	Based on Next-gen HBM
	Rubin Ultra	GPU	AI training	2H27	1TB HBM4e	4 reticles leading to 16 HBM chips each with 64GB memory (based on 16-hi layer 32Gb DRAM)
	Rubin	GPU	AI training	2H26	288/384GB HBM4	HBM4 with 12-hi/16-hi layer 24Gb DRAM die; bandwidth at 13TB/s
	B300	GPU	AI training	3Q25	288GB HBM3e	Based on 8 units of 12-hi layer 24Gb DRAM die; bandwidth at 8TB/s
	B100/200	GPU	AI training	1Q25	192GB HBM3e	Nvidia confirmed production ramp-up and shipment in FY4Q25 (Oct-Jan)
	H200 SXM/NVL	GPU	AI training	2Q24	141GB HBM3e	First GPU with HBM3e (1.2TB/s) memory
	H100 NVL	GPU	AI training	Sep-22	188GB HBM3; 7.8TB/s	Powerful AI training GPU using 188GB HBM3
	H100 SXM5	GPU	AI training	Sep-22	80GB HBM3; 3.35TB/s	High-speed AI training GPU (900GB/sec) PCIe / NVL (600GB/sec)
	H100 PCIe	GPU	AI training	Sep-22	80GB HBM2e; 2TB/s	Still based on HBM2e - HBM3 not yet
	A100	GPU	AI training	Jun-20	40/80GB HBM2/2e; 2TB/s	Old gen but still actively used for AI training
Superchip	Rubin Ultra NVL576	Giant Superchip	AI training/inference	2H27	288TB HBM4e (likely) + 72TB DRAM	Combines 288x Rubin Ultra GPU via NV Link
	Vera Rubin NVL144 CPX	Giant Superchip	AI training/inference	2H26	40TB HBM4 (likely) + 36TB DRAM + 18TB GDDR7	Combines 144x Rubin GPU and 114x CPX GPU and Vera CPUs via NV Link
	Vera Rubin NVL144	Giant Superchip	AI training/inference	2H26	40TB HBM4 (likely) + 36TB DRAM	Combines 144x Rubin GPU along with Vera CPUs via NV Link
	Vera Rubin NVL72	Giant Superchip	AI training/inference	2H26	21TB HBM4 (likely) + 27/54TB DRAM	Combines 72x Rubin GPU along with 36x Vera CPUs via NV Link
	Vera Rubin	Superchip	AI training/inference	2H26	576GB HBM4 + 1.5TB LPDDR5X	Combines 2 x Rubin GPUs with a 1 x Vera CPU
	GB300 NVL72	Giant Superchip	AI training/inference	3Q25	20TB HBM3e + 18TB LPDDR5X	Combines 36x GB300 superchips via NV Link
	GB300	Superchip	AI training/inference	3Q25	576GB HBM3e + 1.2TB LPDDR5X	Combines 2 x B300 GPUs with a 1 x Grace CPU
	GB200 NVL72	Giant Superchip	AI training/inference	1H25	13.5TB HBM3e + 17TB LPDDR5X	Combines 36x GB200 superchips to form one supercomputer
	GB200	Superchip	AI training/inference	1H25	384GB HBM3e + 480GB LPDDR5X	Combines 2 x B200 GPUs with a 1 x Grace CPU
	GH200 v2	Superchip	AI training/inference	2Q24	144GB HBM3e	2nd gen of GH200 using "HBM3e (1.2TB/sec)"
	GH200	Superchip	AI training/inference	Dec-23	96GB HBM3	Superchip based on both GPU+HBM3 and CPU+LPDDR5
China	B30/B40	GPU	AI training/inference	2H25	GDDR7	Despec version of B300 for China; multiple B30 chips can be connected to form cluster
	H20	GPU	AI training	1Q24	96GB HBM3; 4TB/s	Despec version of H100 for China export;
	L2	GPU	AI training/inference	Jan-24	24GB GDDR6; 300GB/s	Despec version of L40S for China export; lower bandwidth vs L20 [
	L20	GPU	AI training/inference	Dec-23	48GB GDDR6; 864GB/s	Despec version of L40S for China export;
	L40S	GPU	AI training/inference	Oct-23	48GB GDDR6; 864GB/s	No HBM, just GDDR6 used but 1.2x/1.7x better AI inference/training vs A100

Source: Company reports, media reports, BofA Global Research

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Exhibit 28: Lots of GPU line-ups for AI training/inference; LPDDR5 and GDDR6 are also used for AI inferencing but we newly see upbeat demand for 12-hi HBM3e from new ASICs (e.g., Google's TPU v7/TPU 8i/8t); AMD MI400 series; Amazon Trainium chips expected to use HBM4

AI accelerators that use HBM memory

Company	Name	Type	AI function	Launch date	DRAM spec	Remarks
AMD	MI500	APU	AI training	2027	500-600GB HBM4e (exp.)	Provides up to a 1,000x increase in performance vs MI300X
	MI450	APU	AI training	2026	432GB HBM4; 20TB/s+	Fabricated on TSMC's 2nm technology; can be scaled to a sever rack combining 72x GPUs (to a total of 30TB HBM)
	MI400	APU	AI training	2026	384GB HBM4	To compete with NVIDIA's Rubin GPU using HBM4; based on 16-hi
	MI350X	APU	AI training	2H25	288GB HBM3e; 8TB/s	Announced during its Advancing AI 2024 event; MP in 2H25
	MI325X	APU	AI training	4Q24	256GB HBM3e; 6TB/s	256GB HBM3e stated during its Advancing AI 2024 event on 10 Oct
	MI300A	APU	AI training	Dec-23	128GB HBM3; 5.3TB/s	Slightly higher bandwidth vs MI300X
	MI300X	APU	AI training	Jun-23	192GB HBM3; 5.2 TB/s	More advanced version vs MI300
	MI300	APU	AI training	Jan-23	128GB HBM3; 1.2TB/s	Combination of CPU+GPU
Intel	Crescent Island	GPU	AI inference	2H26	160GB LPDDR5X	Xe3P architecture with 640-bit memory interface
	Gaudi3	GPU	AI training	3Q24	128GB HBM2e; 3.7TB/s	Target to compete with NVIDIA H100
	Gaudi2	GPU	AI training	May-22	96GB HBM2e; 2.45TB/s	Stated stronger than NVIDIA A100
	Gaudi	GPU	AI training	NA	32GB HBM2; 1TB/s	
Meta	MTIA 400	ASIC	AI inference	2027	288GB HBM; 9.2TB/s	On track to deploy chips in datacenters; 6 petaflops FP8 perf.
	MTIA 300	ASIC	AI inference	2026	216GB HBM; 6.1TB/s	Offers 1.2 petaflops of FP8 compute performance
Amazon	Trainium 4	ASIC	AI training	4Q26/1Q27	288GB HBM4	2x the HBM capacity and 4x the bandwidth (19.6 TB/s)
	Trainium3	ASIC	AI training	4Q25	144GB HBM3e	2x faster and 40% more energy-efficient vs previous gen
	Trainium2	ASIC	AI training	Dec-2024	96GB HBM3e	Apple reportedly will use these chips for training its AI Intelligence
	Inferentia2	ASIC	AI inference	2H22	32GB HBM2e	More advanced AI inference vs 2019 generation
	Trainium	ASIC	AI training	2H22	32GB HBM2e; 1.6TB/s	Used for training AI models in cloud
Google	TPU 8t	ASIC	AI training	2H26	216GB HBM3e	Can be scaled to 9,600 chips to a massive 2PB HBM memory
	TPU 8i	ASIC	AI inference	2H26	288GB HBM3e	Scalable to 1,152 chips for 331TB HBM memory
	TPU v7 (Ironwood)	ASIC	AI inference	4Q25	192GB HBM3e	Designed for inference workloads and for AI agents
	TPU v6e	ASIC	AI training	Late-2024	64GB HBM3	2x higher capacity and bandwidth vs its predecessor
	TPU v5p	ASIC	AI training	Dec-23	95GB HBM2e	2.8x faster in AI training vs its predecessor
	TPU v5e	ASIC	AI training	Aug-23	16GB HBM2	Pre-version of v5 2024; combines tens of thousands of v5e TPUs
Microsoft	Maia 200 (next-gen)	ASIC	AI training	2026	288GB HBM3e	For large-scale training and inference tasks within Microsoft cloud
	Maia 100	ASIC	AI inference	2024	64GB HBM2e	Optimized for Azure/Bing inference workloads;1.8TB/s bandwidth
	Cobalt CPU	CPU	AI inference	TBC	DDR5	Undisclosed
Huawei	Ascend 950PR/950DT	NPU	AI Inference/training	2026	128GB HiBL/144GB HiZQ	Employs proprietary HBM (HiBL / HiZQ)
	Ascend 910C	NPU	AI Training	1H25	128GB HBM2e (likely)	ASIC for China AI training and inference
	Ascend 910B	NPU	AI Training	Oct-23	64GB HBM2e	Mostly used Ai chips in China besides the US chips
Alibaba	Zhenwu M890	PPU	AI training/inference	2026	144GB HBM3	Mainly for Alibaba AI training/inference; supports cluster scaling
Biren	BR106	GPU	AI training/inference	2023	64GB HBM2e	HBM3 not yet adopted; memory content also low at 64GB
	BR104	GPU	AI training/inference	2022	32GB HBM2e	TSMC 7nm process
Enflame	L600	ASIC	AI training	2025	144GB HBM2e	Mainly for China AI training
Cambricon	MLU590	ASIC	AI inference	2024	64GB HBM2e	Mainly for AI inference;

Source: Company reports, media reports, BofA Global Research

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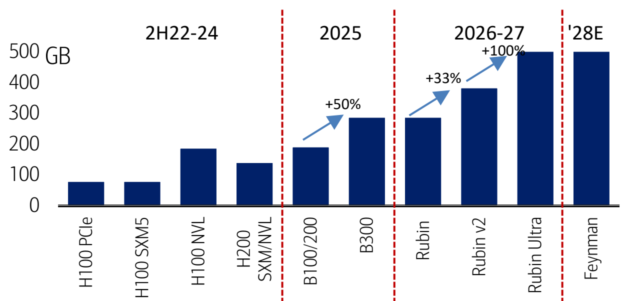
Exhibit 29: Substantial HBM content growth expected in 2025/2026-27, led by B300, Rubin, and Rubin Ultra

NVIDIA GPUs HBM spec evolution; Rubin Ultra HBM content also up 3x vs 2026 1st Rubin



Rubin Ultra and Feynman are likely to have HBM content more than 1TB; axis scaled

Source: NVIDIA, BofA Global Research

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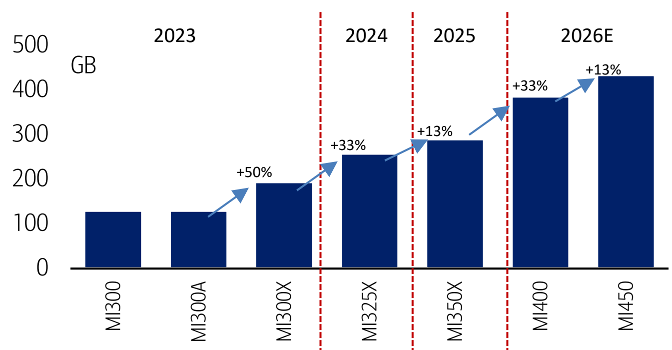
Exhibit 30: 10-30% HBM content growth observed in AMD GPUs as well led by MI350X in 2025 and MI400 series in 2026

AMD GPUs HBM spec evolution



Source: NVIDIA, BofA Global Research

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Exhibit 31: HBM capacity to hit 1TB with Rubin Ultra (2027) and then more than 1.5TB with Feynman Ultra (2030); bandwidth also set to increase 40-50% every year from 1.2 TB/s in 2025 to more than 8 TB/s in 2030
NVIDIA GPU HBM capacity and bandwidth progression

NVIDIA GPU HBM capacity and bandwidth progression

GPU	Shipment start	HBM gen	HBM units/GPU	# of DRAM die layers	DRAM die capacity	Capacity/HBM	Total HBM/GPU	Bandwidth/HBM	GPU bandwidth	Packaging	Bonding
Blackwell B200	2024	HBM3e	8	8-hi	24Gb (3GB)	24GB	192GB	1.2 TB/s	7.2 TB/s	2.5D CoWoS	MR-MUF/NCF
Blackwell Ultra B300	2025	HBM3e	8	12-hi	24Gb (3GB)	36GB	288GB	1.2 TB/s	7-8 TB/s	2.5D CoWoS	MR-MUF/NCF
Rubin	2026	HBM4	8	12-hi/16-hi	24Gb (3GB)	36-48GB	288-384GB	2.8-3.3 TB/s	22-26 TB/s	2.5D CoWoS	MR-MUF/NCF
Rubin Ultra	2027	HBM4e	16	16-hi	32Gb (4GB)	64GB	1,024GB	3.6-4.1 TB/s	58-66 TB/s	2.5D CoWoS	MR-MUF/NCF
Feynman	2028-29	HBM5	8	16-hi/20-hi	32Gb (4GB)	64-80GB	512-640GB	5-6 TB/s	40-50 TB/s	3D Stacked	Partially HB
Feynman Ultra	2030	HBM5e	16	16-hi/20-hi	40Gb (5GB)	80-100GB	1,280-1,600GB	8 TB/s+	128 TB/s+	3D Stacked	Almost Full HB
Post Feynman	2032	HBM6	16	20-hi/24-hi	48Gb (6GB)	120-144GB	1,920-2,304GB	12 TB/s+	192 TB/s+	3D Stacked	HB
Post Feynman Ultra	2035	HBM7	32	20-hi/24-hi	64Gb (8GB)	160-192GB	5,120-6,144GB	20 TB/s+	640 TB/s+	3D Stacked	HB

Source: Company, media reports, BofA Global Research

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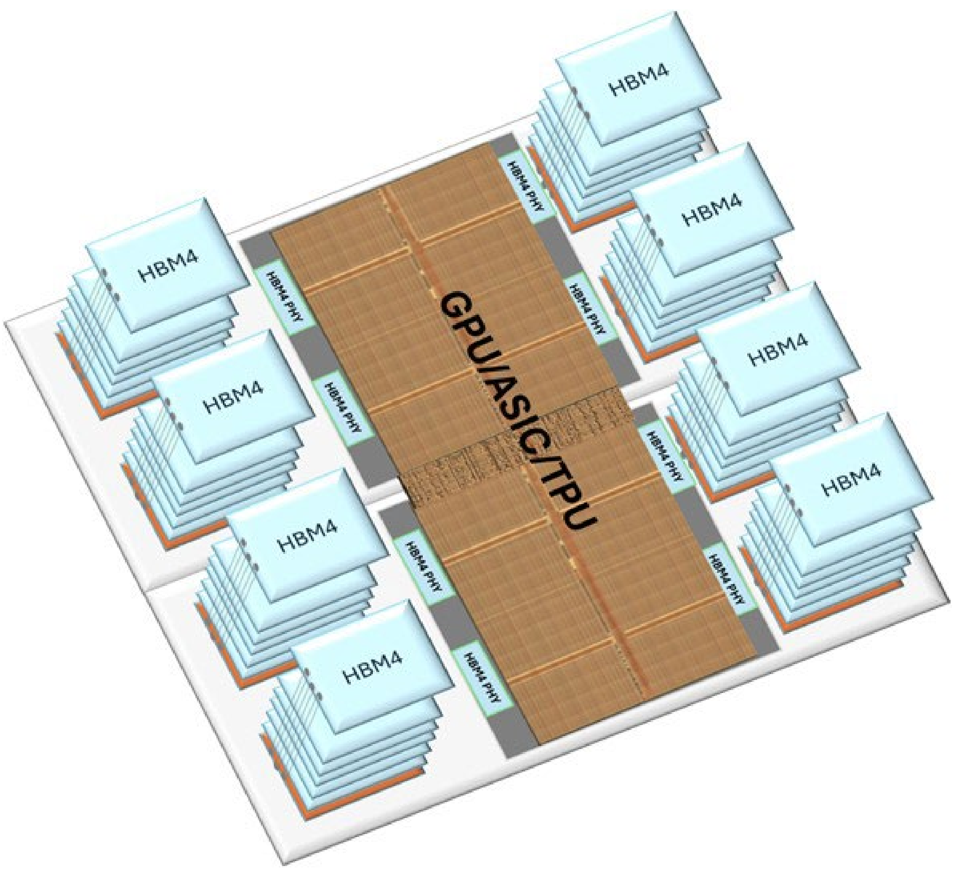
Exhibit 32: GPU/ASIC/TPUs use 6-8 HBM chips; upcoming Rubin Ultra GPU is expected to use 16 HBM units; each HBM contains 8-12 DRAM dies (16-20 likely for HBM4e and HBM5)
HBM supply GPU or ASIC (TPU) vs conventional DRAM (LPDDR5, etc.) works with CPU



Source: BofA Global Research

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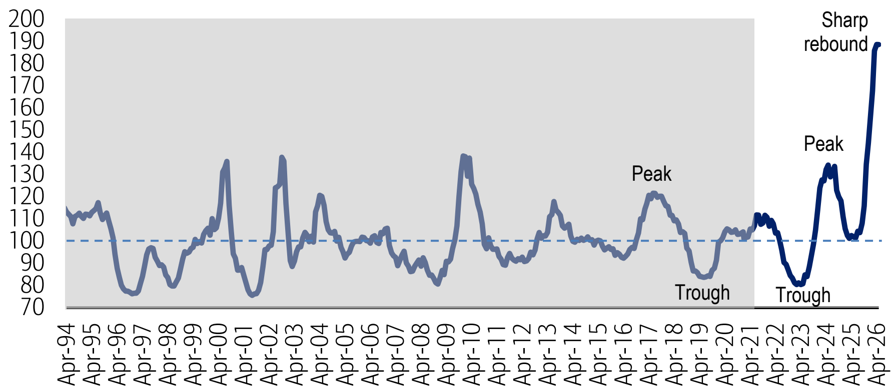
BofA Memory Indicator

Exhibit 33: Our memory indicator reached an all-time high of 189 in Mar/Apr-26, driven by exceptionally strong DRAM/NAND spot pricing, rising ASPs and billings, along with Korea's 150%+ export growth.

BofA Memory Indicator – back to upturn level since Oct-2025 and hit highest peak in Mar/Apr-26 (back-tested)



Note: Our indicator had previously been capped at ~140, but the recent unprecedented surge—driven by memory spot prices, ASPs, and billings—has led us to raise the ceiling (up to 240 levels) to reflect a more accurate relative comparison. This exceptional strength is further validated by strong earnings from memory chipmakers.

Source: DRAMeXchange, WSTS, MoTIR Korea, BofA Global Research

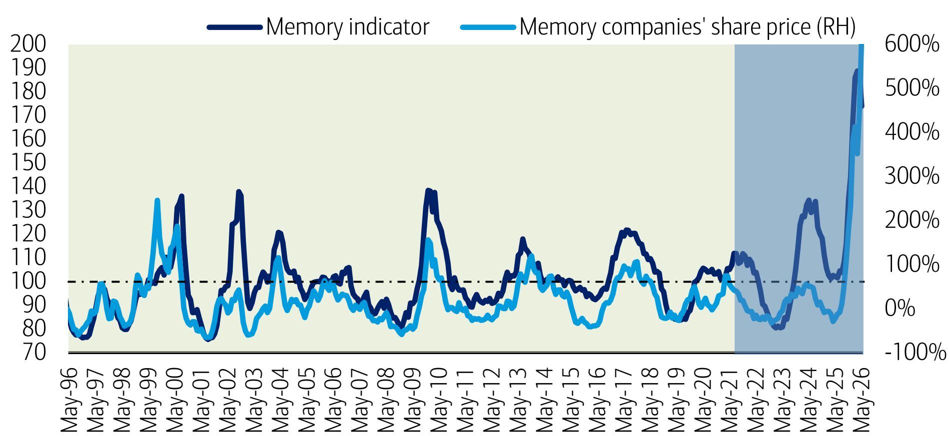
*The shaded area represents back-tested results from January 1991 to March 2021. The unshaded area represents actual performance since April 2021. This performance is back-tested up to March-2021, and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

Disclaimer: The BofA Memory Indicator is intended to be an indicative metric only and may not be used for reference purposes or as a measure of performance for any financial instrument or contract, or otherwise relied upon by third parties for any other purpose, without the prior written consent of BofA Global Research. This indicator was not created to act as a benchmark. For methodology, see the 5 Aug 2024 Global Memory Tech report.

BofA GLOBAL RESEARCH

Exhibit 34: May share price surged to record highs, supported by Samsung (strong 1Q results, optimism around HBM4 upside, and higher exposure to conventional DRAM) and Nanya Tech (record Jan–May monthly sales, up ~500–700% YoY).

BofA Memory Indicator is highly correlated with stock performance (back-tested)



Note: Memory companies share price is average of Samsung, Hynix, Micron, and Nanya share price YoY changes

Source: Bloomberg, BofA Global Research

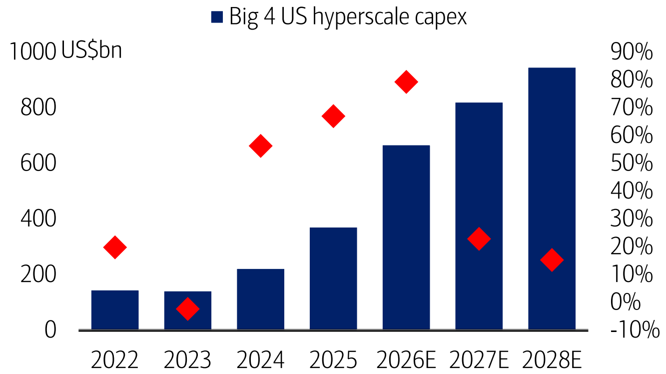
The light green shaded area represents back-tested results from January 1991 to March 2021. The blue shaded area represents actual data since April 2021. This performance is back-tested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein. For methodology, see the 5 Aug 2024 Global Memory Tech report.

BofA GLOBAL RESEARCH

Hyperscale capex and cloud revenue margins

Exhibit 35: Amazon, Microsoft, Alphabet, Meta, and Oracle capex collectively to grow ~80% YoY in 2026 to $650bn even after strong 65% YoY growth in 2025; 2027/28 capex to hit $800/950bn+

Combined capex of top 4 US tech companies

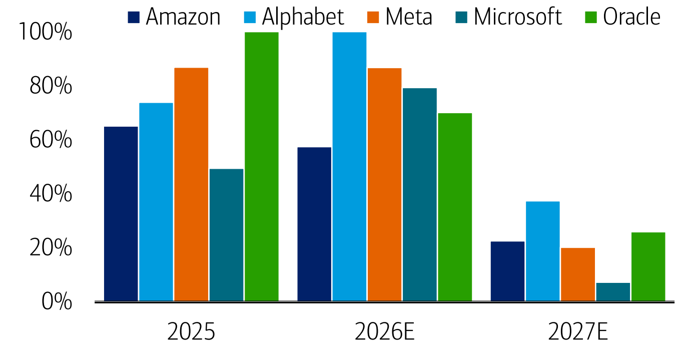


Source: Company reports, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 36: Record-high capex spend by all big-4 hyperscalers expected to continue in 2026E; robust growth since 2024

Hyperscalers' capex growth – YoY change



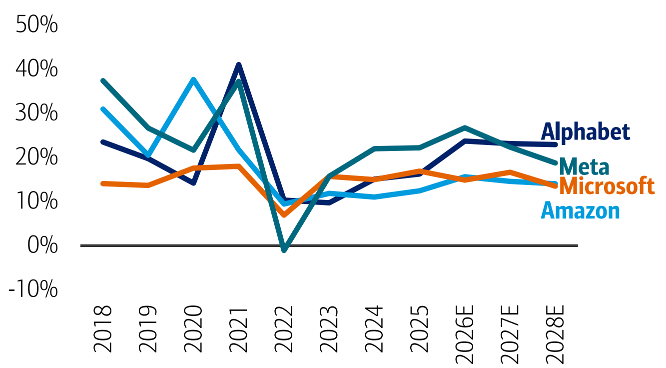
Note: Microsoft and Oracle data adjusted for CY basis **Oracle's 2025 capex up 230% YoY MSFT capex for 2026-27 is based on consensus

Source: Companies, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 37: Amazon, Microsoft, Alphabet, and Meta overall revenue expected to grow +15-20% in 2026-28

Overall revenue growth trend of top 4 US hyperscalers



Source: Company reports, BofA Global Research estimates

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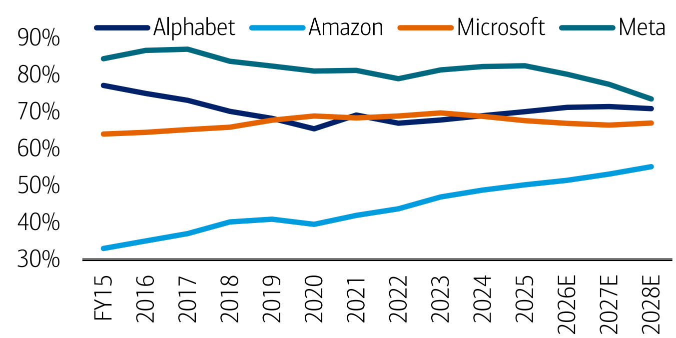
Exhibit 38: Gross margin of hyperscalers strong at around 70-80% levels

Gross margin trend of top 4 US hyperscalers



Source: Company reports, BofA Global Research estimates

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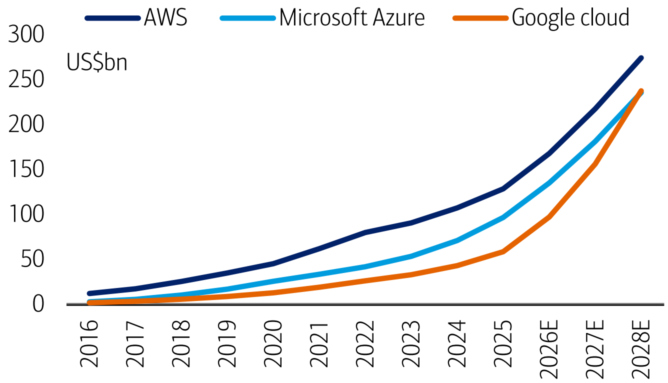
Exhibit 39: Cloud revenue growth still solid at 35-40% YoY in 2026-28
Cloud revenue of hyperscalers consistently strong



Source: Company reports, BofA Global Research estimates

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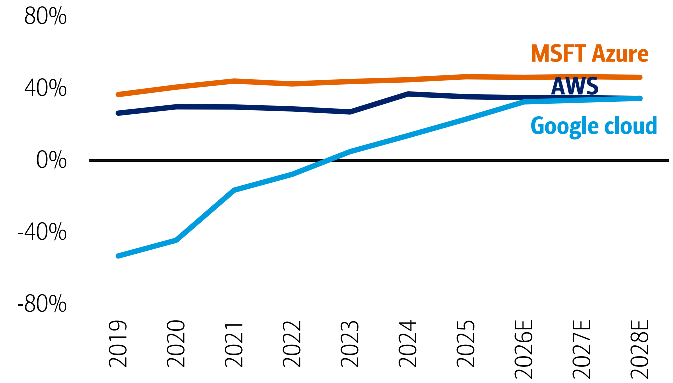
Exhibit 40: High OPM seen among AWS (35%+) and Azure (40%+); Google's margin also set to reach 30-35% in 2026-28
Cloud OPM trend also high

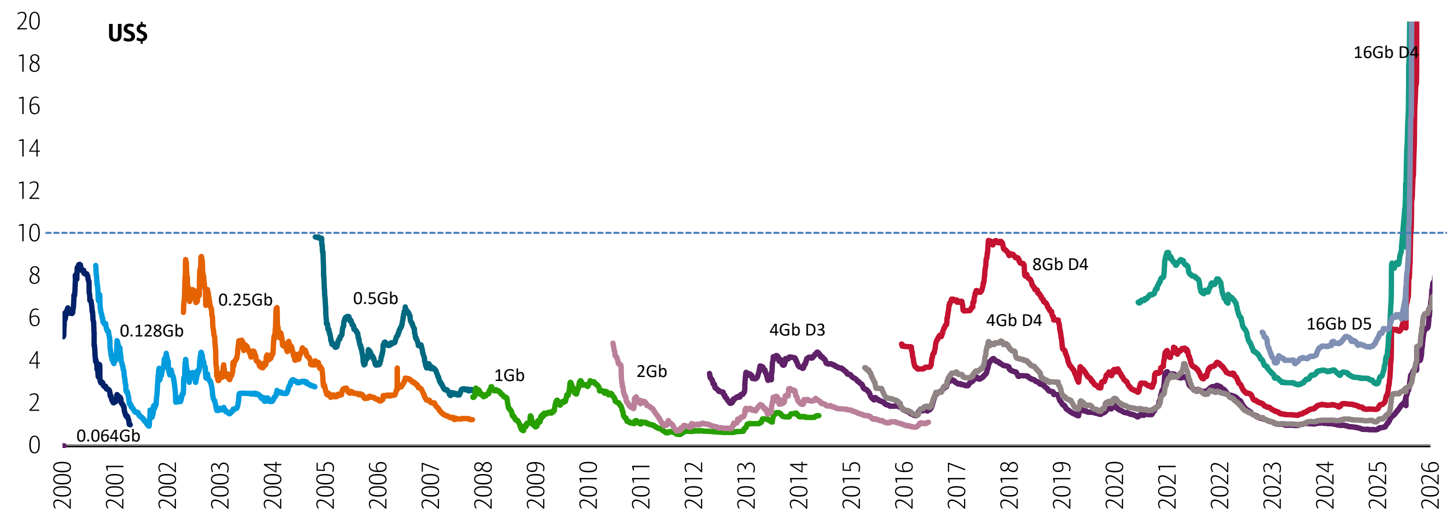


Source: Company reports, BofA Global Research estimates

BofA GLOBAL RESEARCH

Memory spot/contract price trend

Exhibit 41: DRAM spot prices rebounded through June after softening through April–May, following a strong rally from September 2025 to January 2026. Prices have reached multi-decade highs, with 16Gb DDR5 at ~$47 and DDR4 at ~$72, driven by supply reallocation toward HBM and server DRAM—well above the prior ~$10 peak in October 2017.
DRAM spot price – long-term trend (2000-2025)



Source: DRAMeXchange

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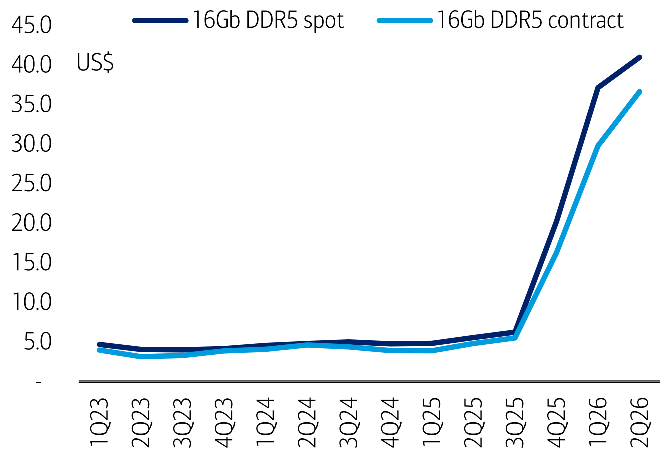
Exhibit 42: DRAM spot and contract prices climbed to record highs of ~$35–40, with moderate growth in 2Q26 following a strong rally in 4Q25 and 1Q26.

DRAM spot and contract price - quarterly average trend

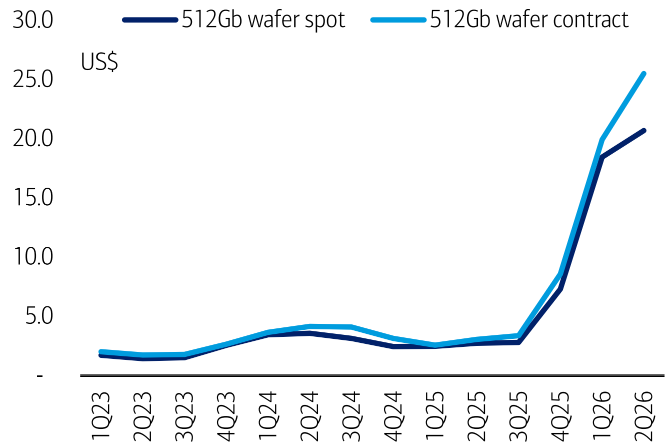


Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

Exhibit 43: NAND wafer contract prices have surged significantly ahead of spot prices, with both now at record-high levels

NAND wafer spot and contract price - quarterly average trend



Source: DRAMeXchange, TrendForce

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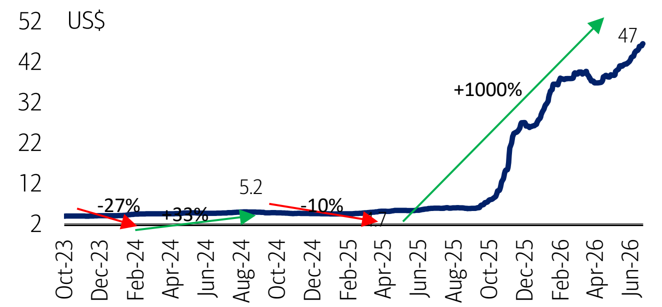
Exhibit 44: Prices rebounded through May and June, reaching an all-time high ($47) following a volatile April. Uptrend was driven by strong gains in Oct (+70%), Nov (+60%), Jan (+25%), before moderating into a more subdued trend during Feb–Mar

16Gb DDR5 spot price – daily, Oct '23 - Jun '26



Source: DRAMeXchange

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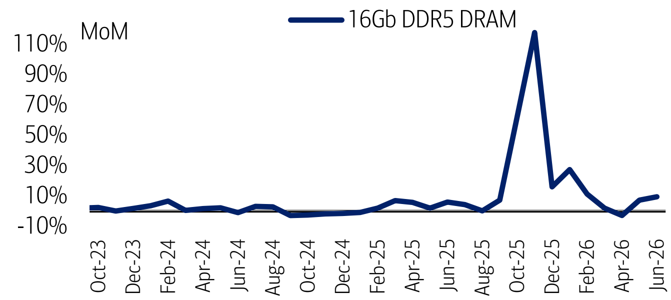
Exhibit 45: Modest uptick observed in May/Jun (+7-10%), following a sequential slowdown from Feb (+10%) to flat momentum in Mar and a minor decline in Apr, after the strong Jan (+25%) surge and the sharp Oct–Nov-25 rally (+60–120%)

16Gb DDR5 spot month-average price – MoM change, Oct '23 - Jun '26

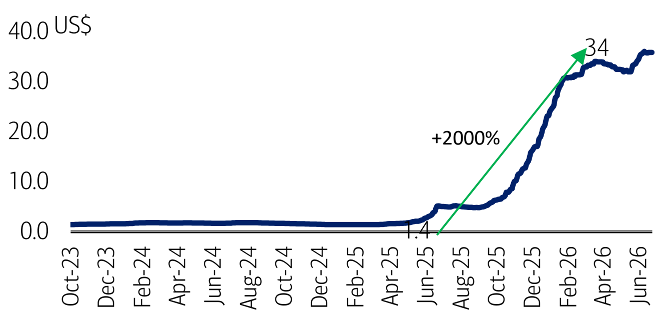


Source: DRAMeXchange

BofA GLOBAL RESEARCH

Exhibit 46: 8Gb DDR4 prices remained flat in 2H-Jun vs up in 1H-Jun and through May after stabilizing in April; they remain significantly higher YoY due to supply cuts by major vendors, currently around $30—well above the prior ~$10 peak in October 2017

8Gb DDR4 spot price trend, Oct '23 - Jun '26



Source: DRAMeXchange

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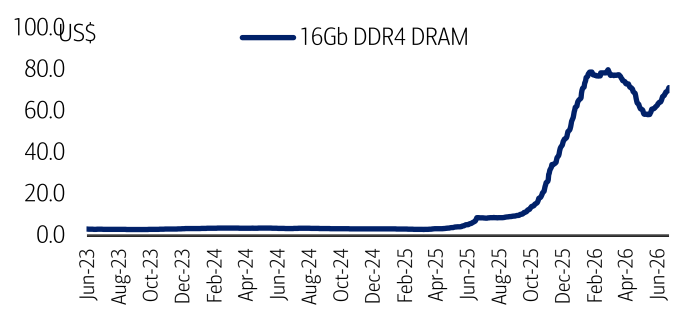
Exhibit 47: Prices edged up slightly in late May/through June after a ~25% correction from April to mid-May, having peaked near ~$80 in early March. Despite this pullback, prices remain elevated—up ~2000% from ~$3 in October 2025—supported by strong gains through Sep–Dec (+30–70%) and January (+20%)

16Gb DDR4 spot price trend, Jun '23 - Jun '26



Source: DRAMeXchange

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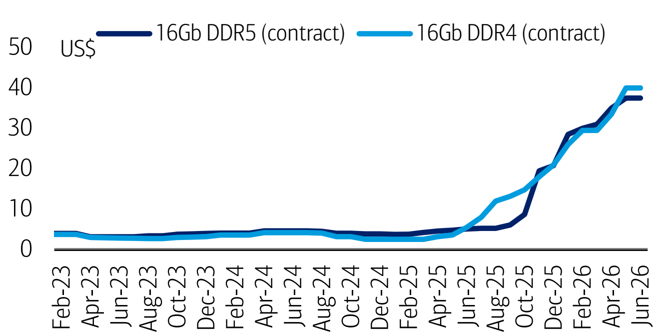
Exhibit 48: DDR4 and DDR5 (16Gb) contract prices are similar at US$35-$40 levels – DDR5 price premium no longer exists due to DDR4 shortage

16Gb DDR5 vs 16Gb DDR4 contract price trend, Feb '23-Jun '26



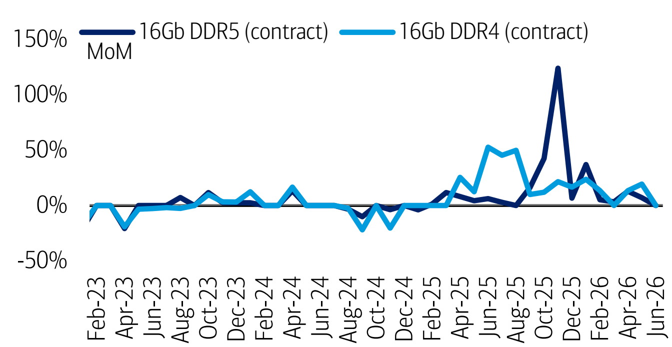
June-26 is an estimate from TrendForce released on 16th June

Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 49: DDR5 and DDR4 contract prices expected to be flat MoM in Jun vs up +10-15% MoM each in Apr/May-26; price strong through 2H25 and early 2026

16Gb DDR5 vs 16Gb DDR4 contract price change - MoM, Feb '23-Jun '26

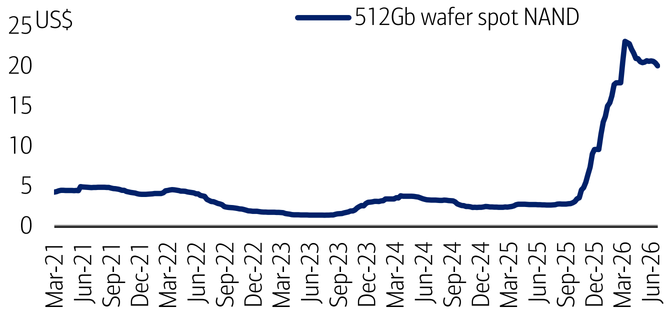


June-26 is an estimate from TrendForce released on 16th June

Source: TrendForce

BofA GLOBAL RESEARCH

Exhibit 50: NAND spot prices broadly stable in 2H-May and softened through-June, after trending down gradually through late March, April, and 1H-May, still up 50%+ YTD and 8x vs Feb-25 low ($2.4)
512Gb NAND wafer spot price – weekly, Mar '21 - Jun '26



Source: DRAMeXchange

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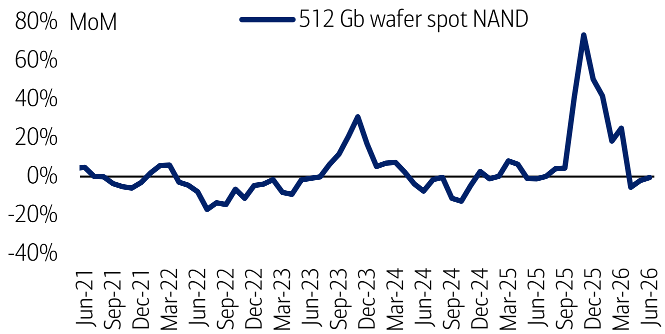
Exhibit 51: Flat in June, while slightly corrected in Apr/May vs up +15-20% in Feb/ Mar-26 and up +40-70% MoM in Oct/Nov/Dec/Jan
512Gb NAND wafer spot average – MoM change, Jun'21 - Jun'26



Source: DRAMeXchange

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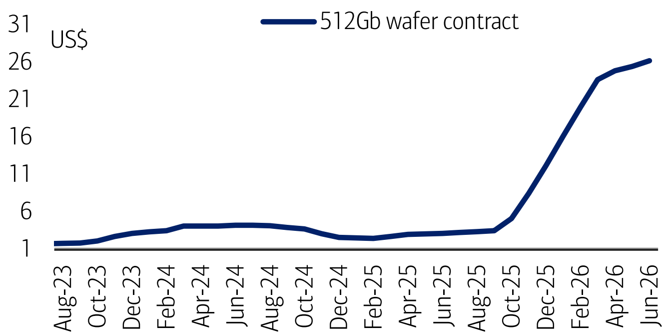
Exhibit 52: Current contract price at ~$25, around 10x vs Feb-25 bottom of $2.5; prices significantly up in 4Q25/1Q26 and remained broadly stable thereafter
NAND wafer contract price trend, Aug '23-Jun '26



June-26 is an estimate from TrendForce released on 16th June
Source: TrendForce, DRAMeXchange, BofA Global Research

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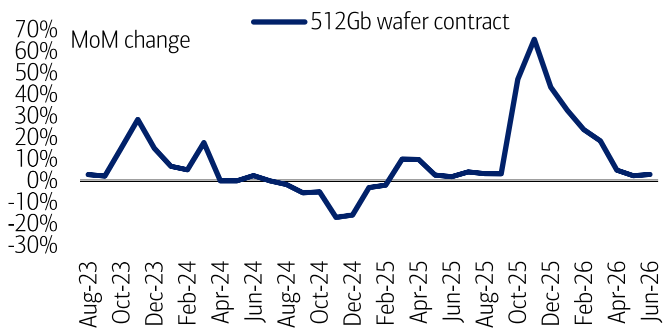
Exhibit 53: Apr/May/Jun price only up +3-5% MoM each following Jan/Feb/Mar-26 rally (up 20-30% MoM) and Oct/Nov/Dec upturn (up 40-60% MoM)
MoM change of NAND contract price (512Gb wafer Aug '23-Jun '26



June-26 is an estimate from TrendForce released on 16th June
Source: DRAMeXchange, TrendForce, BofA Global Research

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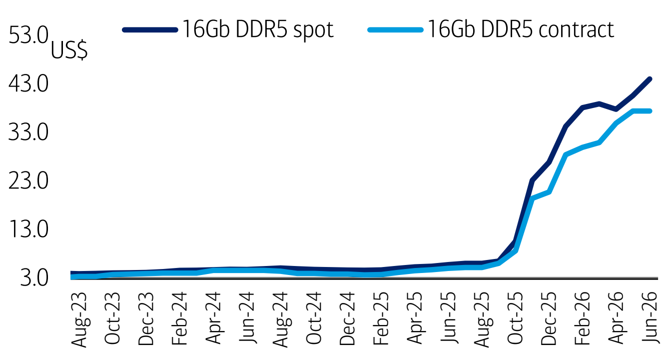
Exhibit 54: 16Gb DDR5 spot and contract prices are in the range of US$35-40 vs historical range of US$3-5
16Gb DDR5 spot vs contract price trend (US$), Aug '23-Jun '26



June-26 is an estimate from TrendForce released on 16th June
Source: DRAMeXchange, TrendForce

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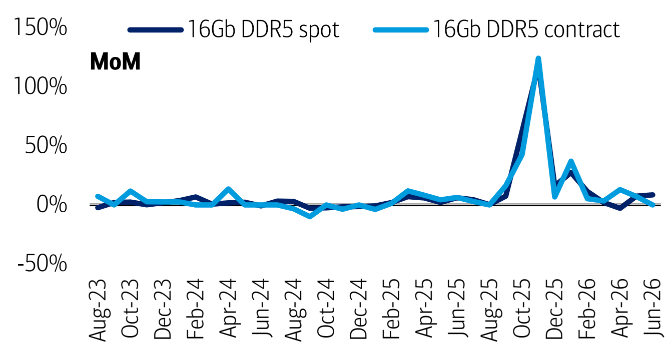
Exhibit 55: Jun'26 spot prices turned positive again while contract price expected to be flat, prices had witnessed a sharp Oct–Dec'25 rally and subsequent Feb–Mar moderation
16Gb DDR5 spot vs contract price trend – MoM change, Aug '23-Jun '26



June-26 is an estimate from TrendForce released on 16th June
Source: DRAMeXchange, TrendForce

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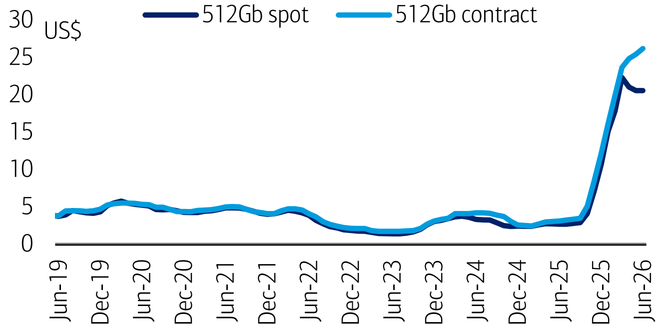
Exhibit 56: Current NAND spot and contract prices are a few times higher than 2025 summer level

512Gb NAND wafer spot vs contract price trend (US$), Jun '19 - Jun '26



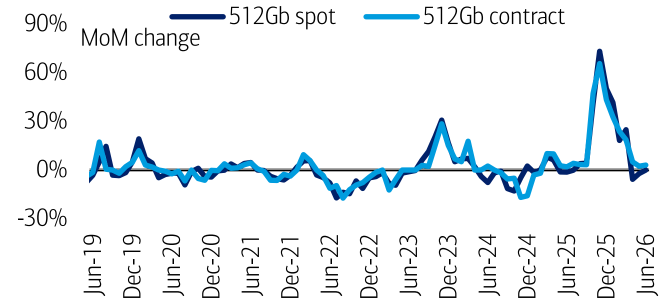
June-26 is an estimate from TrendForce released on 16th June

Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

Exhibit 57: NAND spot and contract prices normalized in Apr/May/Jun after strong rally during Oct-25 – Dec-25 (up 40-50% per month) and Jan-Mar-26 (20-30% per month)

512Gb NAND spot vs contract price trend – MoM change, Jun '19 - Jun '26



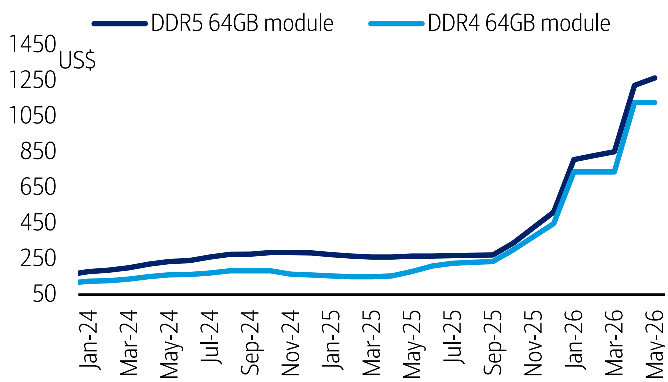
June-26 is an estimate from TrendForce released on 16th June

Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

Exhibit 58: Current price of 64GB DDR4/DDR5 modules hit all-time high of more than $1,000 level (DDR5: US$1,200, DDR4: US$1,100)

Server DRAM contract price trend – DDR5 vs DDR4 modules, Jan '24-May '26



Source: TrendForce

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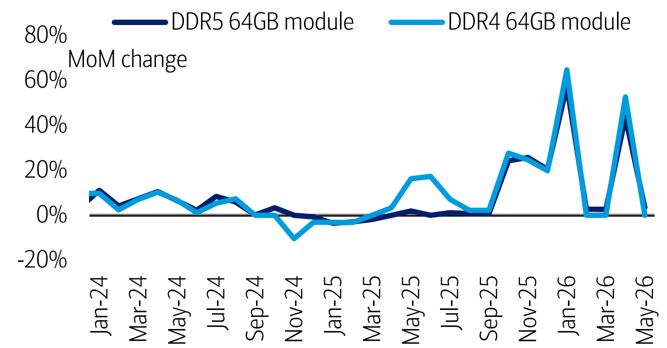
Exhibit 59: DDR4/DDR5 server contract prices remained flat in May, following a sharp 40–50% MoM increase in April 2026 and a similar 50–60% surge in January, after largely stable pricing conditions during February–March

MoM change of server DRAM contract prices, Jan '24-May '26



Source: TrendForce

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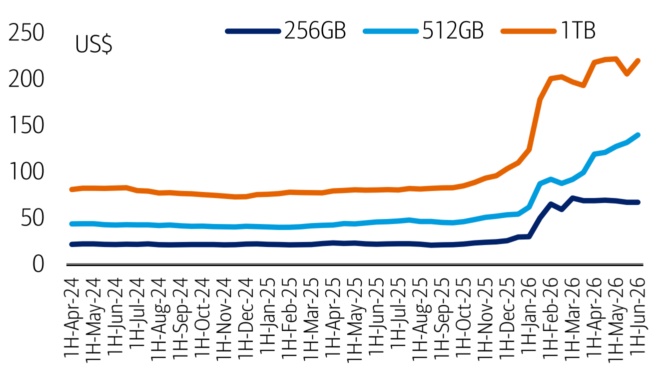
Exhibit 60: 1H-Jun SSD price sharply up vs Mar decline; prices witnessed a sharp surge in 1Q26 and April (per DRAMeXchange), following a gradual upward trend throughout 2025

Client SSD price trend – mostly for PC (not server), Mar '24 - May '26



Note: SSD prices are as of 12 Jun 2026, reported by DRAMeXchange. TB = terabyte.

Source: DRAMeXchange

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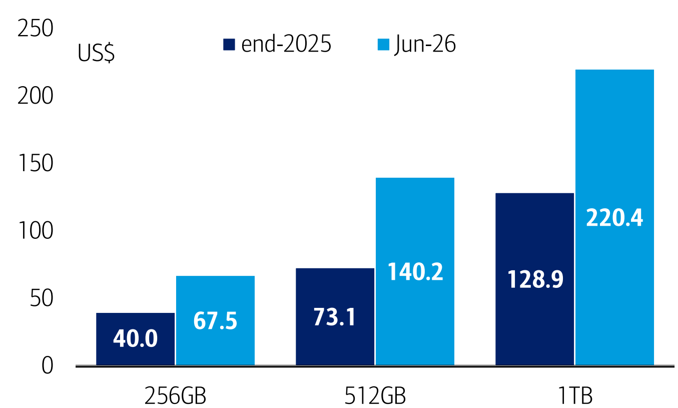
Exhibit 61: Jun prices have doubled compared to end-2025 levels, while the 2025 increase was more modest at around 35–40%

Client SSD (for PC) price comparison – current vs. end-2025



Note: SSD prices are as of 12 Jun 2026, reported by DRAMeXchange

Source: DRAMeXchange

BofA GLOBAL RESEARCH

Valuation and stock performance

Exhibit 62: Memory companies still show very low P/E multiples despite robust stock price rally in May/through-June vs very strong earnings momentum (exceptionally strong DRAM and NAND ASP in 1Q/2Q26)

Valuation comparison among memory and semiconductor supply chain stocks

	Ticker	Rating	Price (Local)	Mcap ($bn)	P/E			P/Book			EV/EBITDA		EV/Sales		ROE			Div. yield	
					FY25	FY26E	FY27E	FY25	FY26E	FY27E	FY26E	FY27E	FY26E	FY27E	FY25	FY26E	FY27E	FY26E	FY27E
Memory/DRAM																			
SK Hynix	HXSCF	RSTR	2,653,000	1,353.1	44.7	-	-	16.0	-	-	-	-	-	-	44.2%	-	-	-	-
Samsung	SSNLF	B-1-7	339,500	1,359.0	55.4	9.4	7.6	5.8	3.7	2.6	5.8	4.7	3.0	2.5	10.8%	47.0%	39.1%	0.8%	1.3%
Micron	MU	C-1-7	1,048.51	1,182.4	126.5	14.3	7.5	21.8	8.7	4.5	11.5	6.1	8.9	5.0	18.8%	87.8%	81.0%	0.1%	0.1%
Nanya	NNYAF	B-1-7	477.00	51.8	216.8	7.4	6.1	9.3	4.5	2.9	5.4	4.4	4.1	3.1	3.9%	73.5%	51.8%	2.6%	3.3%
Memory/NAND																			
Kioxia	XIYUF	C-1-9	103,850	351.2	101.5	11.3	8.7	40.2	8.8	4.4	7.2	5.6	5.8	4.6	51.9%	128.2%	67.2%	0.0%	0.0%
Sandisk	SNDK	C-1-9	1,914.46	283.5	n/a	27.9	10.2	30.8	15.1	5.7	21.4	7.6	13.7	6.3	4.3%	75.8%	86.6%	0.0%	0.0%
Western Digital	WDC	C-1-7	643.83	235.6	n/a	65.2	39.0	41.8	18.0	12.6	41.3	25.0	17.3	12.8	22.5%	42.7%	42.2%	0.1%	0.1%
SIMO	SIMO	C-1-7	321.66	10.9	88.4	35.7	30.2	12.8	9.8	7.9	26.6	20.9	6.2	5.3	15.3%	31.9%	29.6%	0.7%	0.9%
Phison	PISNF	C-1-8	2,475.00	17.2	62.4	6.9	11.7	9.0	4.6	3.8	5.2	8.3	2.1	1.9	16.0%	89.2%	36.0%	7.7%	4.6%
US/Taiwan semis																			
NVIDIA	NVDA	C-1-7	199.00	4,815.8	43.7	21.9	15.0	31.5	16.4	10.3	18.3	12.7	12.4	8.7	94.3%	95.9%	80.3%	0.4%	0.6%
AMD	AMD	C-1-9	519.74	847.5	124.6	69.3	39.2	13.5	11.4	8.8	50.0	30.1	17.0	11.0	11.3%	18.1%	26.3%	0.0%	0.0%
Broadcom	AVGO	C-1-7	382.07	1,817.7	56.0	32.9	21.3	22.5	17.3	11.1	26.1	16.7	17.4	11.4	45.3%	61.3%	65.5%	0.7%	0.7%
Intel	INTC	C-1-9	131.65	661.7	306.2	124.2	76.5	5.2	4.8	4.4	32.8	24.9	11.0	9.5	1.7%	4.1%	6.2%	0.0%	0.0%
Arm	ARM	C-2-9	359.08	383.5	202.9	173.5	126.4	46.3	36.6	28.3	137.1	98.7	64.7	48.3	25.0%	23.6%	25.5%	0.0%	0.0%
Qualcomm	QCOM	B-3-7	197.41	208.1	16.4	18.6	18.4	9.8	7.9	7.6	15.3	15.9	5.3	5.3	56.0%	47.4%	40.6%	1.8%	1.8%
TSMC	TSMWF	B-1-7	2,390	1,949.4	36.1	23.3	15.9	11.4	8.1	5.6	15.0	10.6	11.0	7.9	35.4%	40.8%	41.8%	1.0%	1.0%
MediaTek	MDTKF	B-1-7	4,310	217.4	65.1	66.3	31.6	17.3	18.3	12.6	51.9	25.6	10.5	6.3	26.4%	26.6%	46.8%	1.2%	2.5%
Equipment																			
ASML	ASMLF	B-1-7	1,553.80	684.0	62.7	46.8	33.2	30.8	25.0	20.6	37.1	27.0	14.9	11.9	50.6%	58.7%	67.5%	0.6%	0.7%
AMAT	AMAT	B-1-7	588.97	467.6	62.5	48.5	36.3	22.9	16.0	11.2	41.4	31.0	14.2	11.3	38.6%	39.0%	36.0%	0.4%	0.4%
Lam	LRCX	C-1-7	374.80	468.7	90.8	66.3	49.6	47.5	41.0	29.5	54.5	40.7	20.3	15.9	57.9%	66.9%	68.9%	0.3%	0.3%
TEL	TOELF	C-1-7	75,340	217.9	60.2	50.5	41.0	16.7	14.2	11.8	34.3	28.2	11.0	9.4	29.3%	30.3%	31.4%	1.0%	1.2%
Hanmi Semi	HNSIF	C-1-7	273,500	17.0	179.3	128.2	67.0	55.8	42.1	27.9	100.0	49.9	49.0	27.3	34.8%	37.3%	49.8%	0.3%	0.5%
BESI	BESVF	B-1-7	292.90	27.0	173.8	67.9	40.6	55.5	40.6	29.7	51.7	32.4	23.4	16.6	29.2%	69.4%	84.6%	1.4%	2.4%
ASMPT	ASMVF	C-1-7	204.80	11.0	80.4	51.2	30.9	5.0	4.7	4.3	29.2	19.1	4.4	3.1	6.5%	9.4%	14.5%	1.0%	1.3%

Source: Company reports, BofA Global Research estimates; Mcap = market capitalization, ROE = return on equity, Div = dividend. RSTR = Restricted. Price as of latest close.

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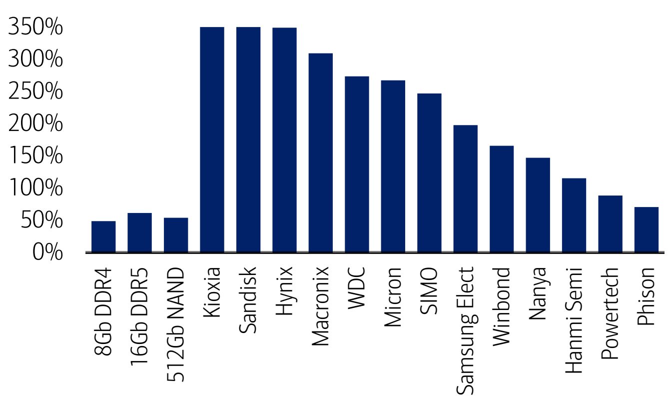
Exhibit 63: Memory prices strongly up this week following Micron's upbeat results/guidance; prices remain elevated year-to-date, led by NAND and HDD with sustained strength in DRAM, underpinned by robust AI-driven demand

Memory companies – 2026 YTD stock performance comparison



SanDisk/Kioxia shows above 800% YTD return

Source: Bloomberg, DRAMeXchange

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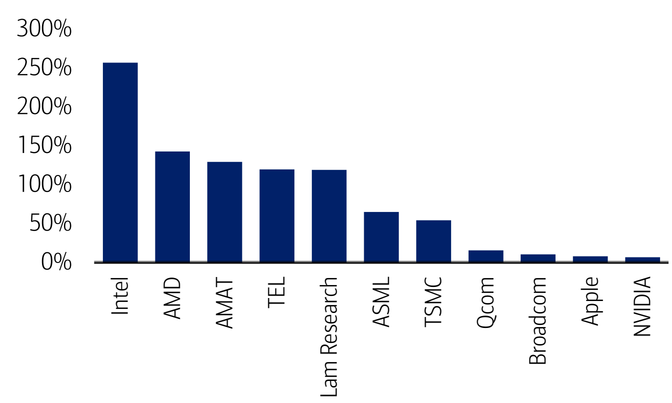
Exhibit 64: CPU names such as Intel and AMD clearly outperforming other US big tech names such as NVIDIA/Apple/QCOM

Global major tech companies – 2026 YTD stock performance comparison



Source: Bloomberg

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Glossary

AI: Artificial intelligence
ASP: Average selling price
avg: average
chg: change
DDR4/5: 4th/5th gen double-data rate DRAM
DRAM: Dynamic random-access memory
eSSD: Enterprise solid-state drive
EUV: Extreme ultraviolet lithography
GDDR7: 7th gen graphics DRAM
GM: Gross margin
HBM: High bandwidth memory
HBM4/4e: 6th/7th gen of HBM
LPDDR5: Low-power DDR5
LTA: Long-term agreement
OEM: Original equipment manufacturer
OPM: Operating profit margin
shr: share
SOCAMM: Small-outline compression attached memory module

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